Contemporanous Correlation Estimation with Mean Shifts

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Abstract

Correlation is a foundational statistical quantity that underlies many statistical methods. Unfortunately, in the presence of mean shifts, classical methods for estimating correlation are biased, and their hypothesis tests are spurious. We introduce the *equivariant covariance estimator* (ECE), which produces unbiased covariance estimates in the presence of mean shifts. We also study a more general framework of unbiased quadratic estimators and show that under the natural assumption of *equivariance*, ECE is unique in the most complex and flexible model class. We also derive a corresponding hypothesis test for the presence of contemporaneous correlation. To our knowledge, ECE is the first method to estimate and test for contemporaneous correlation in this setting. Our approach addresses a key gap in the literature and has implications for econometrics and other disciplines where contemporaneous correlation and structural breaks are both prevalent.

1 Introduction

Correlation is one of the most important statistical quantities because it is the simplest concept of association. This is especially important in multivariate time series, where there are many types of dependencies. For example, since data is collected successively, temporal dependencies (e.g., autocorrelation and cross-correlation) naturally occur. While these correlations play a prominent role in time series, in this report, we restrict our attention to cross-sectional (contemporaneous) correlation: the instantaneous association between time series. Mentions of correlation will henceforth be assumed to be contemporaneous unless otherwise stated.

Time series are an abundant data type that can be found in nearly every field: finance [Cochrane, 2001, Tsay, 2005], economics [Hsiao, 2007], environmental science [Sicard et al., 2023, Sorooshian et al., 2024, Zhu et al., 2019], physics [Ying et al., 2009, Song and Russell, 1999], biology [Zhang, 2010, Wang et al., 2021], engineering [Zhang et al., 2026], and more. Given its widespread applications, it is unsurprising that cross-sectional dependencies are commonplace. Environmental scientists often track air pollution across several sites, where geographically similar sites observe similar observations and are thus spatially dependent. Even within a site, there may be associations between pollutants due to atmospheric chemistry [Sorooshian et al., 2024]. In a similar way, economists study panel data, where economies may be dependent by geography, but they may also be dependent due to historical or political reasons (Elhorst, 2014).

Understanding these dependencies can be insightful. For example, financial analysts use the covariance matrix to model a portfolio's diversity and volatility [Cochrane, 2001]. Alternatively, neuroscientists use the covariance matrix to understand how brain regions interact with one another [Mohanty et al., 2020]. While the covariance matrix may provide some insight, it is also an essential component for inference.

Many models assume the data is cross-sectionally independent (add source), and using these models on data that violates this assumption results in spurious inference [Hsiao, 2007]. To defend against this type of model misspecification, researchers have devised numerous tests for cross-sectional correlation [Breusch and Pagan, 1980, Frees, 1995, Pesaran, 2015, Chudik et al., 2011]. If the data is cross-sectionally correlated, the covariance matrix is used to adjust the inference using methods such as feasible generalized least squares (FGLS) [Parks, 1967], panel-corrected standard errors (PCSE) [Beck and Katz, 1995], seemingly unrelated regression (Zellner, 1962), or common correlated effects [Pesaran, 2006].

As the big data revolution evolves, contemporaneous correlation will play an even larger role. Improved data collection capabilities have produced new sources of data, and these sources must be aggregated and processed. For example, wearable devices can monitor several physiological signals simultaneously, which can be used to predict health outcomes [Seneviratne et al., 2017]. Several fields, such as manufacturing and agriculture, are experiencing a renaissance due to the burgeoning integration of Internet of Things (IoT) technologies. Sensors continuously stream environmental data such as soil moisture, temperature, and light intensity, which can be used to detect diseased crops and improve crop yields (Savita, 2023). In the same way, manufacturers can monitor the health of their equipment, which can reduce the downtime of their equipment and improve the overall quality of their products (Liang, 2024; Tao, 2018).

Researchers have developed many methods to address the challenges posed by big data – many rely on contemporaneous covariance. Principal component analysis (PCA) is used to reduce the dimensionality of time series, where the principal components are derived by estimating the covariance matrix and applying a eigendecomposition on it [Jollife, 2002]. Graphical models are a popular exploratory data analysis tool, but their construction relies on estimating the inverse of the covariance matrix [Epskamp et al., 2018]. Causal discovery methods can uncover the causal skeleton of the data and improve model robustness and interpretability, but they rely on conditional independence tests, which require the calculation of the covariance matrix [Hasan et al., 2024].

Unfortunately, many of the aforementioned methods assume the data is *stationary*: the data-generating process does not change over time. This is a serious problem since time series data are rife with non-stationarity. They may exhibit trends, seasonality, or abrupt deviations from their historical behavior. Stock data trends upwards over the course of decades [Tsay, 2005]. Ozone levels vary seasonally; they are higher in the summer than in the winter [Sicard et al., 2023, Sorooshian et al., 2024, Zhu et al., 2019]. Financial and economic time series may experience structural breaks via policy shifts, financial crises, or global pandemics. As a reflection on the pervasiveness of the issue, the literature is rich with methods that incorporate structural breaks into their models [Bai, 2010, Hansen, 2001, Parsaeian, 2024, Casini and Perron, 2018, Baltagi et al., 2016, Barigozzi et al., 2018]. Failure to account for these non-stationarities produces spurious correlations, which can impact downstream analysis [Onatski and Wang, 2021].

Of course, since non-stationary time series are commonplace, practitioners have devised a variety of approaches to mitigate their complications. Most methods transform the data into a stationary distribution, thereby making classical methods viable. For example, if the data is a random walk, the *first difference* can be applied to obtain the stationary changes between observations [Hsiao, 2007]. Alternatively, if the data exhibits deterministic trends, the trend may be estimated and subtracted, so the remaining residuals will be stationary. Consider the case where the data has a mean that is piecewise constant. The piecewise mean can be estimated by first estimating the change points, and then taking the sample mean of each segment [Xiong et al., 2015]. Alternatively, smoothing methods (e.g., moving medians) can be used to estimate the mean [Nilsen et al., 2012]. More sophisticated modeling techniques, such as Single Spectrum Analysis (SSA) and wavelet analysis, can also be employed to remove trends [Hassani, 2021, Alexandrov et al., 2012]. However, authors have noted that there is no consensus on how to model trends in time series [Min Qi and Zhang, 2008].

Significant effort has been poured into adapting popular methods to accommodate non-stationary data. For example, many variations of non-stationary PCA have been proposed [De Roover et al., 2014, Hamilton et al., 2024, Lansangan and Barrios, 2009]. Similarly, the network modeling and causal discovery literature has proposed methods to circumvent non-stationarity [Basu and Subba Rao, 2023, Mameche et al., 2025].

Despite the rich set of solutions for non-stationarity, challenges remain. For instance, multivariate change point methods often require an estimate of the contemporaneous covariance matrix [Wang and Samworth, 2018]. This is challenging because estimating the covariance matrix in the presence of mean shifts requires an estimate of change points. In other words, to estimate change points, we must estimate the covariance matrix, which requires estimating change points. This is a circular logic that lacks a good solution. Many authors sidestep the issue entirely by assuming there is no contemporaneous correlation at all [Grundy et al., 2020, Zhang et al., 2010]. Others, who acknowledge its existence require assumptions on its extent and must produce cumbersome theoretical justifications [Bai, 2010, Wang and Samworth, 2018].

Moreover, each of these solutions requires input from the analyst. This may be feasible for small dimensions, but as the dimension of the data continues to grow, it becomes increasingly challenging for analysts to directly remove trends from *every* time series. If non-parametric smoothing techniques are applied, the analyst must determine the tuning parameters. If change point detection is applied, the analyst must select the algorithm and penalty type [van den Burg and Williams, 2022]. Applying first differencing may not make the data stationary; it might require a second differencing. While database managers can provide suggestions for appropriate transformations [McCracken and Ng, 2015], having dedicated maintainers is infeasible for most datasets.

It is likewise equally unscaleable to rely on researchers to devise specialized techniques to handle non-stationarity [De Roover et al., 2014, Hamilton et al., 2024, Basu and Subba Rao, 2023, Mameche et al., 2025]. What the literature demands is a non-stationary alternative to Pearson correlation. This would enable the simple substitution of classical methods with a non-stationary alternative. In this report, we make strides in this direction.

The report is organized as follows. We begin by describing the model assumptions in Section 2. After establishing the model, we motivate our proposed estimator and establish its theoretical properties in Section 3.1. In Section 4, we elaborate on the method and propose a hypothesis testing framework. Finally, in Section 5 we compare our method against alternatives in simulated and real data.

2 Model Assumptions

This section describes the data-generating model used throughout the report. For simplicity, we present a Gaussian version, although the same estimation ideas apply more broadly.

Consider a sequence of independent random vectors

$$Z_i = \begin{bmatrix} X_i \\ Y_i \end{bmatrix} \sim \mathrm{BVN}(\theta_i, \Sigma), \qquad i = 1, \ldots, n,$$

where θ_i is the mean vector and Σ is the covariance matrix:

$$\theta_i = \begin{bmatrix} \theta_{x,i} \\ \theta_{y,i} \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{xy} & \sigma_y^2 \end{bmatrix}.$$

It is helpful to view each observation as signal plus noise:

$$Z_i = \theta_i + \varepsilon_i,$$

with

$$\varepsilon_i = \begin{bmatrix} \varepsilon_{x,i} \\ \varepsilon_{y,i} \end{bmatrix} \sim \mathrm{BVN}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma \right), \qquad i = 1, \ldots, n.$$

Because $\{\varepsilon_i\}_{i=1}^n$ are independent across i, there is no serial dependence:

$$\mathrm{Cov}(\varepsilon_{x,i}, \varepsilon_{x,j}) = \mathrm{Cov}(\varepsilon_{x,i}, \varepsilon_{y,j}) = 0, \qquad i \neq j.$$

At the same time, contemporaneous dependence is allowed through $\sigma_{xy} = \mathrm{Cov}(\varepsilon_{x,i}, \varepsilon_{y,i})$, which is the main quantity of interest in this report.

For the mean structure, let $\theta = (\theta_1, \ldots, \theta_n)^T$ and assume it is piecewise constant. We call i a change point when $\theta_i \neq \theta_{i+1}$, and write all change points as $\tau = (\tau_1, \ldots, \tau_J)^T$ in ascending order. Using $\tau_0 = 0$ and $\tau_{J+1} = n$, we summarize the shortest segment length by

$$\mathcal{L}(\theta) = \min_{0 \leq j \leq J} \{\tau_{j+1} - \tau_j\}, \qquad \Theta_L = \{\theta : \mathcal{L}(\theta) \geq L\}.$$

This framework allows frequent mean shifts while still requiring a minimal spacing between them. In what follows, we focus on $L = 2$ and $L = 4$.

3 Covariance Estimation

We begin with the classical covariance estimator and then explain why it fails under the mean-shift setting introduced in Section 2.

With independent and identically distributed (IID) observations $\{(X_i, Y_i)\}_{i=1}^n$, the covariance

$$\sigma_{xy} = \mathrm{Cov}(X, Y) = \mathbb{E}\left((X - \mathbb{E}X)(Y - \mathbb{E}Y)\right)$$

is commonly estimated by the *sample covariance*,

$$\hat{s}_{xy} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}), \tag{1}$$

which estimates the population means $\mathbb{E}X$ and $\mathbb{E}Y$ with sample means $\bar{X}$ and $\bar{Y}$. This works when the mean is constant but it breaks down when the mean changes over time, because global averages cannot time-varying means.

A common workaround is to estimate the mean first and then compute covariance from the residuals. This strategy can work in simple settings, but it becomes unreliable when the mean structure is complex, such as the one described in Section 2.

Our approach avoids explicit mean estimation and instead estimates covariance directly from local differences. We first develop the idea in a simplified setting and then return to the full mean-shift model. We close the section with its main theoretical properties.

3.1 Circular Covariance Estimation

The sample covariance estimator in (1) is biased in the presence of mean shifts because it uses a *constant* sample mean $(\bar{X}, \bar{Y})$ to estimate a *non-constant* population mean $(\mathbb{E}X, \mathbb{E}Y)$. A natural alternative is to use lagged differences, which compare observations locally rather than globally. Define

$$T_k^\ell(X, Y) = \sum_{i=1}^{n-k} (X_i - X_{i+k})(Y_i - Y_{i+k}), \tag{2}$$

and the corresponding *k-lagged covariance estimator*

$$\hat{\sigma}_k^\ell(X, Y) = \frac{1}{2(n-k)} T_k^\ell(X, Y). \tag{3}$$

When there are no mean shifts, $\hat{\sigma}_k^\ell(X, Y)$ is unbiased for σ_{xy} for any $1 \le k < n$. Because this holds across multiple lags, it suggests using several lags together can improve efficiency.

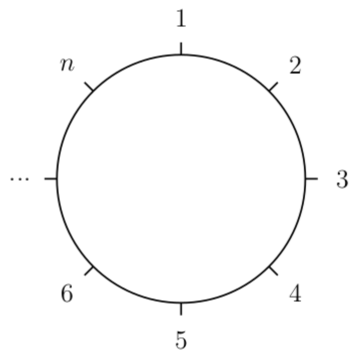


Figure 1: Equivariant embedding.

In (2), the lagged cross-products terms $(X_i - X_{i+k})(Y_i - Y_{i+k})$ are only defined for $i \le n-k$, so the number of lagged cross-products decreases as k grows. To keep the sample size fixed across lags, we embed the index set on a circle so the first and last indices are neighbors

(Figure 1). More formally, we redefine the indexing

$$i := [(i-1) \mod n] + 1. \tag{4}$$

Under this new indexing scheme, the circular lagged cross-product sum is

$$T_k(X,Y) = \sum_{i=1}^{n} (X_i - X_{i+k})(Y_i - Y_{i+k}) \tag{5}$$

and the corresponding circular covariance estimator is

$$\hat{\sigma}_k(X,Y) = \frac{1}{2n} T_k(X,Y). \tag{6}$$

We abbreviate $T_k(X) := T_k(X,X)$ and $T_k(Y) := T_k(Y,Y)$, and $\hat{\sigma}_k(X) := \hat{\sigma}_k(X,X)$ and $\hat{\sigma}_k(Y) := \hat{\sigma}_k(Y,Y)$.

Like the non-circular estimator, this estimator is unbiased for $1 \leq k < n$ in the setting without mean shifts. It is also slightly more efficient than the non-circular version because it uses n lagged differences for every k.

3.2 Equivariant Covariance Estimation

We have established that the circular covariance estimator is unbiased in the absence of mean shifts. Returning to the full model with mean shifts, however, the estimator is biased. The diagonal terms σ_x^2 and σ_y^2 can be estimated by the equivariant variance estimator (EVE) of Hao et al. [2022], which uses the same debiasing approach. We generalize this method to the off-diagonal term σ_{xy} and establish its key properties.

Proposition 1 (Bias of k-Lagged Circular Estimators) *Let $k \in \{1, ..., L\}$, where L is the minimum segment length between change points. Then, the expected value of the lagged difference estimator is*

$$\mathbb{E}\left[T_k(X,Y)\right] = 2n\sigma_{xy} + k\omega_{xy}. \tag{7}$$

where $\omega_{xy} = T_1(\theta_x, \theta_y)$.

It follows from Proposition 1 that

$$\mathbb{E}\left(\hat{\sigma}_k(X,Y)\right) = \sigma_{xy} + k\frac{\omega_{xy}}{2n}. \tag{8}$$

When mean shifts are present, $\hat{\sigma}_k$ is biased, with the bias growing linearly in k. This structure suggests a regression-based correction. Treating $\hat{\sigma}_k(X,Y)$ as the response and k as the predictor, equation (8) defines a line with intercept σ_{xy} and slope $\frac{\omega_{xy}}{2n}$. Regressing on k therefore recovers σ_{xy} as the intercept estimate. This scheme is illustrated in Figure 2.

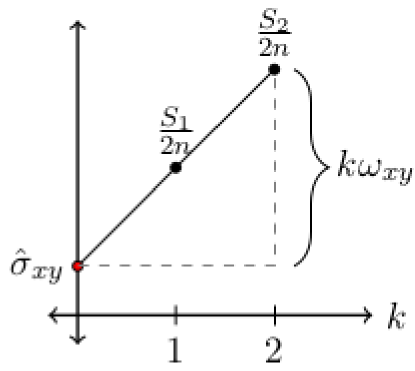


Figure 2: ECE estimator is a regression through $\frac{T_k(X,Y)}{2n}$ for $k = 1, 2$.

More formally, define the design matrix and response vector as

$$K = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad R = \begin{bmatrix} R_1 \\ R_2 \end{bmatrix},$$

where $R_k = \frac{T_k(X,Y)}{2n}$. The first column of K is an intercept term and the second encodes the lag k. Applying ordinary least squares, we obtain

$$(K^T K)^{-1} K^T R = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \end{bmatrix}, \tag{9}$$

giving the intercept estimate $2R_1 - R_2$ and the slope estimate $R_2 - R_1$. The slope estimate yields

$$\hat{\omega}_{xy} = \frac{1}{n}(T_2(X,Y) - T_1(X,Y)), \tag{10}$$

and the intercept defines the *equivariant covariance estimator* (ECE):

$$\hat{\sigma}_{xy} = \frac{1}{2n}\left(2T_1(X,Y) - T_2(X,Y)\right). \tag{11}$$

Setting $X = Y$ recovers the EVE estimator, so ECE is a bivariate generalization of EVE.

Theorem 1 (Unbiasedness of $\hat{\sigma}_{xy}$) *Under the model assumptions described in Sections 2, the estimator defined in (11) satisfies*

$$\mathbb{E}[\hat{\sigma}_{xy}] = \sigma_{xy}.$$

Proof of Theorem 1. This follows easily from (7)

$$\begin{aligned}
\mathbb{E}\left(\hat{\sigma}_{xy}\right) &= \mathbb{E}\left(\frac{1}{2n}(2T_1(X,Y) - T_2(X,Y))\right) \\
&= \frac{1}{n}\left(2n\sigma_{xy} + \omega_{xy} - n\sigma_{xy} - \omega_{xy}\right) \\
&= \sigma_{xy}.
\end{aligned}$$

3.2.1 Bilinear Framework

We have shown that $\hat{\sigma}_{xy}$ is unbiased. To place this in a broader context, we now characterize the full class of bilinear estimators. An estimator $\tilde{s}$ is *bilinear* if $\tilde{s}(X,Y) = X^T AY$ for some $A \in \mathbb{R}^{n\times n}$. This is a natural class of estimators to analyze since the bilinear form encapsulates a set of functions rich enough to contain the population covariance and sample covariance.

We constrain our analysis to bilinear estimators that are also unbiased. First, we rewrite the bilinear estimator $\tilde{s}$ as

$$\begin{aligned}
\tilde{s}(X,Y) &= X^T AY \text{ for } A \in \mathbb{R}^{n\times n} \\
&= (\theta_x + \varepsilon_x)^T A(\theta_y + \varepsilon_y) \\
&= \theta_x^T A\theta_y + \varepsilon_x^T A\theta_y + \varepsilon_y^T A\theta_x + \varepsilon_x^T A\varepsilon_y.
\end{aligned}$$

Taking the expectation, we obtain

$$\mathbb{E}\left(\tilde{s}(X,Y)\right) = \theta_x^T A \theta_y + tr(A)\sigma_{xy}.$$

Therefore, for a bilinear estimator to be unbiased, A must satisfy two properties:

- $\theta_x^T A \theta_y = 0$ for all $\theta_x, \theta_y \in \Theta_L$
- $tr(A) = 1$.

Identifying a matrix A that satisfies these properties is challenging because the search space is so large. We can reduce the complexity of the problem by invoking a natural symmetry property: *equivariance.*

3.2.2 Equivariance

The sample covariance (1) is permutation invariant: its value is unchanged under any reordering of the observations. Under IID assumptions this is appropriate, since the ordering carries no information. However, when the means shift over time, the ordering is informative. We therefore relax permutation invariance to circular shift invariance, requiring only that the estimator be unchanged under circular shifts of the index set. This is a weaker symmetry condition that permits the estimator to exploit local structure. Note that the sample covariance satisfies circular-shift invariance as well, since circular shifts are a special case of permutations.

To formalize this concept, let C_k be the $n \times n$ circulant shift matrix defined by

$$C_{k,ij} = \begin{cases} 1, & j - i = k \mod n \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

Intuitively, $C_k X$ shifts each element of X, k indices to the left. If the shifted index exceeds the number of indices, then it wraps around to the beginning. For example, if $X = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \end{pmatrix}$, then $C_2 X = \begin{pmatrix} 3 & 4 & 5 & 1 & 2 \end{pmatrix}$.

Then, an estimator $\tilde{s}$ is said to be *equivariant* if it is invariant to circular shifts:

$$\tilde{s}(C_k X, C_k Y) = \tilde{s}(X, Y).$$

This fact is relevant to the bilinear framework, because it significantly reduces the search space of A. Instead of finding $A \in \mathbb{R}^{n\times n}$, we reduce the search to finding $A \in \mathbb{R}^n$ through the following lemma.

Lemma 1 (Equivariant-Circulant Relationship) *The bilinear estimator $\tilde{s}(X, Y) = X^T AY$ is equivariant if and only if A is circulant:*

$$A = circ(a_0, ..., a_{n-1}) = \begin{bmatrix} a_0 & a_{n-1} & ... & a_1 \\ a_1 & a_0 & ... & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1} & ... & a_1 & a_0 \end{bmatrix}$$

Proof of Lemma 1.

$$\begin{aligned} \tilde{s}(C_k X, C_k Y) &= (C_k X)^T A(C_k Y) \\ &= X^T (C_k^T A C_k) Y. \end{aligned}$$

Therefore, $\tilde{s}$ is equivariant if and only if $A = C_k^T A C_k$, which implies A is circulant [Fuhrmann, 2012].

With A restricted to circulant matrices, the n^2 unknowns reduce to the n entries $(a_0, \ldots, a_{n-1})$ of the first row. The unbiasedness condition $\theta_x^T A\theta_y = 0$ for all piecewise-constant $\theta \in \Theta_2$ is equivalent to requiring that the sum of entries of A indexed by any block of l consecutive indices equals zero, for $l = 2, \ldots, \lceil n/2 \rceil$. Applying these constraints sequentially, we uniquely determine A.

Theorem 2 ($\hat{\sigma}_{xy}$ is the unique unbiased bilinear equivariant estimator) *Let $\hat{\sigma}_{xy}$ be the estimator defined in (11), and assume the minimum segment length satisfies $L \geq 2$. Consider all bilinear estimators of the form $X^T AY$, where $A \in \mathbb{R}^{n\times n}$. If the estimator* *is unbiased and equivariant, then A is uniquely determined, and $\hat{\sigma}_{xy}$ is the unique such estimator.*

Proof in Appendix.

One can verify directly that the unique matrix A determined above corresponds exactly to the estimator $\hat{\sigma}_{xy}$ defined in (11), confirming that ECE is the unique unbiased equivariant bilinear estimator.

4 Hypothesis Testing

In the previous section, we produced an estimate of $\hat{\sigma}_{xy}$ and derived some of its useful properties. In practice, researchers are often more interested in testing whether cross-sectional correlation exists, rather than producing an estimate. If cross-sectional correlation is present, it may violate key assumptions in common modeling techniques [Hsiao, 2007]. To address this, we consider the following hypothesis test:

$$H_0 : \rho = 0 \quad \text{vs.} \quad H_1 : \rho \neq 0,$$

where $\rho = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$. Since we have unbiased estimates for σ_x^2, σ_y^2, and σ_{xy}, a natural correlation estimate is

$$\hat{\rho} = \frac{\hat{\sigma}_{xy}}{\hat{\sigma}_x \hat{\sigma}_y} = \frac{2T_1(X,Y) - T_2(X,Y)}{\sqrt{(2T_1(X) - T_2(X))(2T_1(Y) - T_2(Y))}}. \tag{13}$$

Our goal is to derive the asymptotic distribution of $\hat{\rho}$ under H_0, so it can be used to construct a valid test. We begin by defining

$$V = (T_1(X), T_2(X), T_1(Y), T_2(Y), T_1(X,Y), T_2(X,Y))^T,$$

and the differentiable function

$$g(v) = \frac{2v_5 - v_6}{\sqrt{(2v_1 - v_2)(2v_3 - v_4)}}, \tag{14}$$

so $\hat{\rho} = g(V)$. Then, if V is asymptotically normal, $\hat{\rho}$ is also asymptotically normal by the multivariate delta method fix this in bibtex[Kiefer, 2006].

To state the result more formally, we make a series of simplifications. First, without loss of generality, we assume the variances are normalized: $\sigma_x^2 = \sigma_y^2 = 1$, which can always be enforced by rescaling the data. Secondly, we will limit the scope of this report to the analysis of the asymptotic variance of $\hat{\rho}$ under the null hypothesis. With these simplifications, we formally show that V is asymptotically normal.

Lemma 2 (Asymptotic Normality of V) *Under a mild regularity condition:* $\max_j \{(\mu_j - \mu_{j+1})^2\} = o(n)$*), we have*

$$\sqrt{n}(V - \mu_V) \xrightarrow{d} N(0, \Sigma_V),$$

Where $\mu_V = \mathbb{E}(V)$ *and* $\Sigma_V = \mathrm{Cov}(V)$.

Proof in Appendix.

The delta method states that $\mathrm{Var}(\hat{\rho}) = \nabla g^T(V)\Sigma_V \nabla g(V)$ asymptotically. It is straightforward to compute that under the null hypothesis, $\nabla g(\mu_V)|_{\rho=0} = (0, 0, 0, 0, 2, -1)^T$. Therefore, for the purposes of deriving the asymptotic distribution of $\hat{\rho}$, we do not need to fully characterize Σ_V – we only need expressions for $\mathrm{Var}(T_k(X, Y))$ and $\mathrm{Cov}(T_1(X, Y), T_2(X, Y))$.

Proposition 2 (Covariance Matrix) *Let* $\kappa_{rs} = \mathbb{E}\left(\varepsilon_x^r \varepsilon_y^s\right)$. *For* $k \in \{1, 2\}$,

$$\mathrm{Var}\left(T_k(X, Y)\right) = 4n\kappa_{22} + 2n(1 - \rho^2) + 2kT_1(\theta_x) + 2kT_1(\theta_y) + 4k\rho T_1(\theta_x, \theta_y), \tag{15}$$

$$\mathrm{Cov}\left(T_1(X, Y), T_2(X, Y)\right) = 4n(\kappa_{22} - \rho^2) + 2T_1(\theta_x) + 2T_1(\theta_y) + 4\rho T_1(\theta_x, \theta_y). \tag{16}$$

Proof in Appendix.

Clearly, under the null hypothesis,

$$\mathrm{Var}(\hat{\rho}) = \nabla g(\mu_V)^T \Sigma_V \nabla g(\mu_V) = 4\mathrm{Var}(T_1(X, Y)) - 2\mathrm{Cov}(T_1(X, Y), T_2(X, Y)) + \mathrm{Var}(T_2(X, Y)).$$

Substituting the values calculated in Proposition 2, we obtain an explicit form for the asymptotic variance under the null.

Theorem 3 (Asymptotic Distribution of $\hat{\rho}$) *Let $\hat{\rho}$ be the estimator defined in Equation 13. Under the regularity conditions described in Lemma 2, it follows under the null hypothesis that*

$$\sqrt{n}\hat{\rho} \xrightarrow{d} N(0, \nu^2).$$

where $\nu^2 = \nabla g(\mu_V)^T \Sigma_V \nabla g(\mu_V) = \omega_x^2 + \omega_y^2 + \frac{7}{2}$.

We see that the asymptotic distribution of $\hat{\rho}$ depends on the norm terms ω_x and ω_y, which reflect the total variation in the mean structure. In practice these are unknown quantities and must be estimated. We recommend using the estimates obtained from the regression scheme described in (10).

5 Numerical Results

5.1 Simulated Data Examples

Having established the asymptotic properties of the equivariant covariance estimator (ECE), we turn our attention towards its finite-sample properties. This allows us to evaluate the estimator's performance in realistic settings and test its robustness to model misspecification. To empirically analyze its inferential utility and efficiency, we establish its control over type I error and estimates its power across a variety of settings.

Each setting is evaluated using Monte Carlo simulations, repeated 10,000 times to produce reliable estimates of type I error and power. Within each replication, we estimate correlation with Equation (13 and produce its corresponding p-value using the asymptotic variance described in Theorem 3. This p-value determines whether we accept or reject the null hypothesis at a $\alpha = 0.05$ significance level. By aggregating this across all replications, we estimate the empirical type I error (when $\rho = 0$) and power (when $\rho > 0$).

In each simulated scenario, we generate our data $X = \theta_x + \varepsilon_x$ and $Y = \theta_y + \varepsilon_y$, where $\varepsilon_x, \varepsilon_y \sim N(0, 1)$, $\rho = Cor(\varepsilon_x, \varepsilon_y)$, and θ_x, θ_y are defined based on the scenarios in Table 1. Scenario 1 represents a setting with a single mean shift (minimal non-stationarity), whereas scenario 2 represents an extreme setting with frequent mean shifts (maximal non-stationarity). To evaluate robustness to model misspecification, scenarios 3 and 4 introduce departures from the piecewise constant model. Scenario 3 exhibits a sinusoidal seasonal trend, while scenario 4 represents a random walk. Finally, scenario 5 represents a realistic setting that still adheres to the piecewise constant model. It is a semi-random walk, where the mean evolves via random increments like an ordinary random walk, but the increments do not occur at every time step. Instead, they occur at random intervals (each of at least length four). In both scenarios 4 and 5, a random seed is fixed to ensure reproducibility across replications. That is, the generated random walks are consistent across all replications.

To benchmark ECE's performance, we compare it against three competing techniques. All techniques apply some transformation to make the data stationary and subsequently estimate the Pearson correlation. The first is the *oracle estimator*, which assumes the change points are known, and estimates θ_x and θ_y using the sample mean of each segment. We subtract the estimated θ_x and θ_y from X and Y, making the data stationary. Finally, we apply Pearson correlation. This estimator serves as a best-case scenario, providing an upper bound on achievable performance.

We will refer to the second technique as *segmentation*, which is similar to the oracle estimator, but assumes the change points are *unknown* and must be estimated. In the literature, there are many methods to detect change points [Xiong et al., 2015, Shi and Touge, 2023, Truong et al., 2018, Niu et al., 2016], but a comprehensive study by van den Burg and Williams [2022] identified the pruned exact linear time (PELT) method [Killick et al., 2012] to be one of the most accurate change point detection procedures. Therefore, the segmentation method uses PELT to identify the change points. We use the default implementation from the `changepoint` R package, but apply the BIC penalty, which is more sensitive to change points than the default MBIC. We also set the minimum segment length to two to avoid degenerate segmentations.

Finally, the last technique we will numerically evaluate is *Segment-ECE*. As in the segmentation method, we use PELT to estimate and remove mean shifts, but we apply ECE instead of Pearson correlation. The rationale is that eliminating large deterministic mean changes should improve ECE's rejection accuracy: since the rejection region depends on the size of $\|\theta\|$, removing large shifts improves the power.

Together, these three procedures form a set of natural benchmarks for evaluating ECE. The oracle estimator represents the best possible performance under perfect knowledge of the change points; the segmentation method reflects what can be achieved when the mean is estimated from the data; and segment-ECE tests whether combining segmentation with ECE improves inference by removing large shifts before estimation. With these baselines established, we now examine how each method behaves in finite samples by comparing their empirical type I error and power across all scenarios.

5.1.1 Type I Error

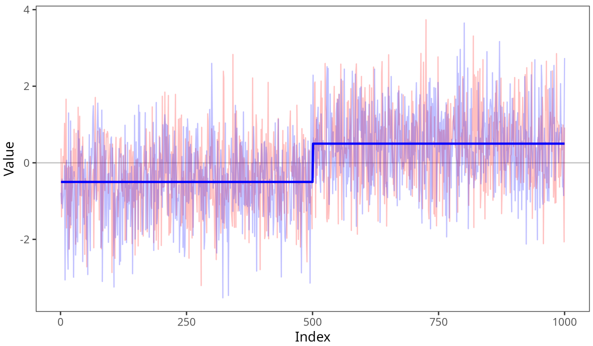
Simulation results for type I error are summarized in Table 2. Unsurprisingly, Pearson correlation fails at controlling type I error because it fails to account for the mean shifts. It conflates the static signal with random variation, thereby consistently and incorrectly rejecting the null hypothesis.

In contrast, by estimating the mean shifts with the segmentation method, we obtain some control over type I error, but only situationally. It has good control when change points are obvious (e.g. scenario 1), but struggles when they are frequent and difficult to detect (e.g. scenarios 2 and 5). Moreover, increasing the sample size does not guarantee improved performance. Instead, in most of the scenarios, increasing the sample size *decreases* the performance. This is because PELT, the change point detection technique underlying segmentation, relies on penalties that adversely scale with sample size. To detect a change point, the fit must improve as a function of the penalty. However, common choices for the penalty (e.g. BIC and MBIC) are functions of n. Therefore, the segmented fit must improve more as sample sizes increase.

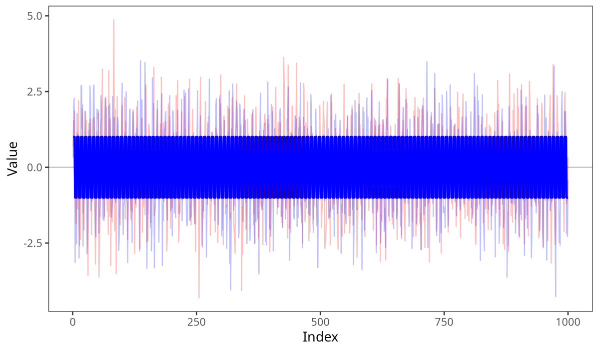
Remarkably, even the oracle estimator, which has exact knowledge of the change point locations, also fails to consistently control for type I error. It particularly fails when the segment lengths are short (e.g. scenarios 2 and 5). Despite knowing the correct segmentation, it is insufficient because the within-segment sample means overfit the data, thus underestimating the within-segment variances (σ_x^2 and σ_y^2). This causes the resulting correlation estimate to have an exaggerated spread, causing excessive false rejections. This is most obvious in

Scenario	Definition	Description
1	$\theta_{x,i} = \theta_{y,i} = \begin{cases} \frac{1}{2}, & 1 \leq i \leq n/2 \\ -\frac{1}{2}, & n/2 < i \leq n \end{cases}$	Single change point at midpoint.
2	$\theta_{x,i} = \theta_{y,i} = \begin{cases} 1, & (i-1) \bmod 8 < 4 \\ -1, & (i-1) \bmod 8 \geq 4 \end{cases}$	Mean shift every four observations.
3	$\theta_{x,i} = \theta_{y,i} = \sin\left(\frac{2\pi}{365} i\right), \quad i = 1, ..., n$	Misspecified model with seasonal trend.
4	$\theta_{x,i} = \sum_{j<i} \varepsilon_{x,j}, \quad i = 1, ..., n$ $\theta_{y,i} = \sum_{j<i} \varepsilon_{y,j}, \quad i = 1, ..., n$	Misspecified model with random walk.
5	$\theta_{x,i} = \sum_{m=1}^{M} \mu_{x,m} \mathbb{1}\{\tau_m < i \leq \tau_{m+1}\}$ $\theta_{y,i} = \sum_{m=1}^{M} \mu_{y,m} \mathbb{1}\{\tau_m < i \leq \tau_{m+1}\}$	Semi-random walk.

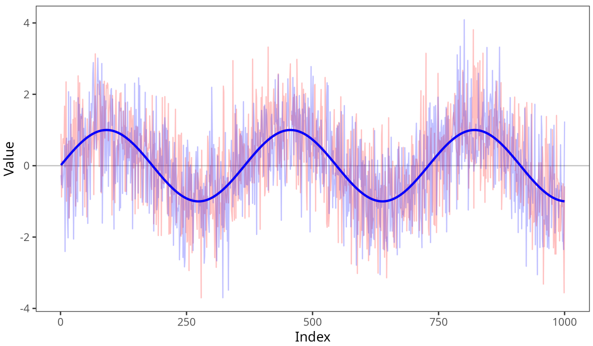
Table 1: Definitions and descriptions of the simulation scenarios. Scenario 5: $M = 0.1n$; $\mu_{x,j}, \mu_{y,j} \sim N(0, 1)$; and change points $\tau_{x,j}, \tau_{y,j}$ are each drawn from $4 \cdot \{1, \ldots, n/4\}$ without replacement.



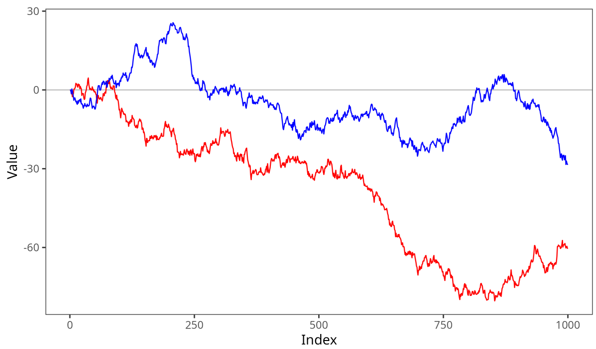
(a) Scenario 1



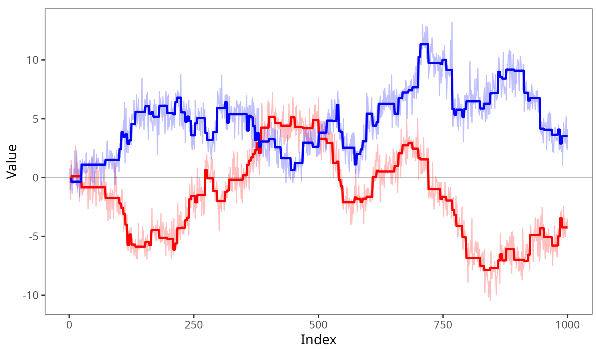
(b) Scenario 2



(c) Scenario 3



(d) Scenario 4



(e) Scenario 5

Figure 3: Simulated data instance for each scenario described in Table 1 with sample size $n = 1000$. Each panel displays one instance of (X, Y) illustrating the underlying mean structure and corresponding data points.

settings with frequent mean-shifts (e.g. scenario 2).

Contrast this with ECE, which nearly achieves the desired significance-level in every scenario. Small sample results show mildly inflated type I errors, but this bias disappears with larger sample sizes, consistent with its asymptotic unbiasedness. Despite this, ECE's type I error control remains stable across both frequent mean shifts and model misspecification, demonstrating that it effectively accounts for non-stationarity. The only situation where it fails is in the case of the random walk, where variance estimates are volatile and may even be negative. Therefore, the corresponding correlation estimates may not produce a real number.

Fortunately, by applying segmentation before ECE, we maintain similar type I errors, with the added benefit that it stabilizes results for the random walk. While its false positive rate is still slightly inflated, it still does better than its counterparts, while also enjoying the robustness of ECE in other scenarios.

Taken together, these results demonstrate that mean-removal strategies cannot reliably achieve type I error control, but ECE and its segmented variant can. Therefore, we limit our power analysis to ECE and segment-ECE in the next section.

5.1.2 Power Analysis

Table 3 summarizes a power analysis of ECE and segment-ECE benchmarked against the oracle estimator where θ is known. The oracle estimator is essentially a Pearson correlation estimate with no mean shifts. Obviously, it is more powerful than the other methods since it does not estimate the mean vector; however, it serves as a useful upper bound on the achievable power.

Relative to the oracle, both ECE and segment-ECE achieve modest power. Moreover, their performance remains relatively stable across all scenarios. Slight differences between scenarios can be attributed to the magnitude of $\|\theta\|$, since ECE's standard error depends on this quantity. As established in Theorem 3, a larger $\|\theta\|$ inflates the standard error, which in turn increases the resulting p-value. Consequently, when mean shifts occur frequently, the test becomes less sensitive, making it more difficult to detect a significant correlation.

We see this more clearly when comparing ECE against segment-ECE. Segment-ECE

n	Method	Scenario 1	Scenario 2	Scenario 3	Scenario 4	Scenario 5
400	ECE	0.057	0.056	0.055	**NA**	0.055
	Segment-ECE	0.055	0.056	0.060	0.063	0.056
	Pearson	**0.982**	**1.000**	**1.000**	**0.882**	**0.982**
	Oracle (τ)	0.049	**0.090**	***	***	0.065
	Segmentation	0.054	**1.000**	0.056	**0.103**	**0.099**
1000	ECE	0.053	0.053	0.052	**NA**	0.051
	Segment-ECE	0.052	0.054	0.053	0.069	0.052
	Pearson	**1.000**	**1.000**	**1.000**	**0.925**	**1.000**
	Oracle (τ)	0.051	**0.088**	***	***	0.063
	Segmentation	0.052	**1.000**	**0.075**	**0.110**	**0.124**
5000	ECE	0.049	0.050	0.054	**NA**	0.050
	Segment-ECE	0.049	0.048	0.049	0.069	0.053
	Pearson	**1.000**	**1.000**	**1.000**	**0.968**	**1.000**
	Oracle (τ)	0.046	**0.093**	***	***	0.062
	Segmentation	0.050	**1.000**	**0.339**	**0.129**	**0.992**

Table 2: Empirical type I error rates of the ECE and Pearson correlation estimators under the five simulation scenarios described in Table 1. The nominal significance level is 0.05. Type I errors that differ from this nominal value are bolded. The oracle estimator cannot be applied to misspecified scenarios, so we fill the entries with ***.

Method	Scenario	Sample Size	Correlation (ρ) 0.05	0.15	0.25
Oracle(θ)		400	0.170	0.850	0.999
		1000	0.348	0.997	1.000
ECE	1	400	0.093	0.383	0.793
		1000	0.139	0.724	0.992
	2	400	0.080	0.291	0.622
		1000	0.115	0.564	0.931
	3	400	0.097	0.380	0.784
		1000	0.139	0.730	0.992
	4	400			
		1000			
	5	400	0.085	0.360	0.755
		1000	0.134	0.693	0.987
Segment-ECE	1	400	0.090	0.378	0.792
		1000	0.137	0.727	0.991
	2	400	0.085	0.308	0.652
		1000	0.119	0.567	0.935
	3	400	0.090	0.381	0.801
		1000	0.143	0.740	0.990
	4	400	0.070	0.114	0.202
		1000	0.076	0.176	0.360
	5	400	0.097	0.391	0.786
		1000	0.159	0.728	0.991

Table 3: Empirical power rates of the ECE under the three simulation scenarios described in Table 1. Each entry is based on repeated simulations with sample sizes $n = 400$ and $n = 1000$. We have also provided the oracle estimator as a benchmark to compare performance.

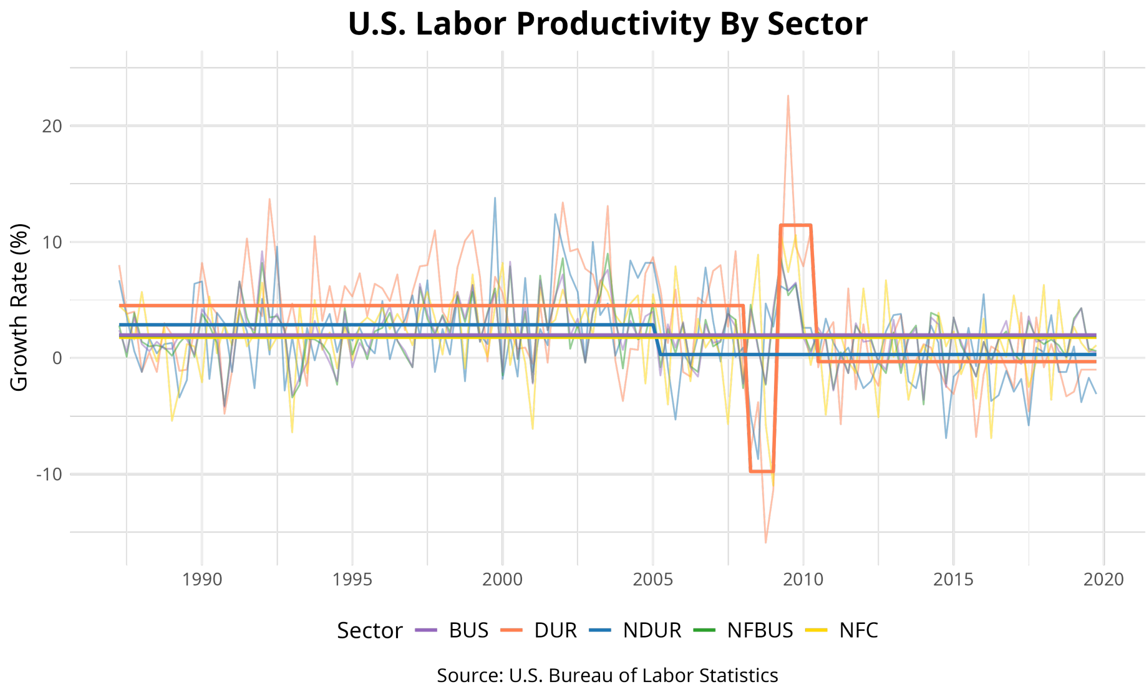


Figure 4: Volatility of labor productivity across durable (DUR) and nondurable (NDUR) manufacturing sectors. Data from the U.S. Bureau of Labor Statistics.

estimates the mean vector and subtracts it, thereby reducing the size of $\|\theta\|$. This, in turn, shrinks the standard error of the subsequent ECE estimate, boosting the sensitivity to detect correlations. This is why segment-ECE has improved power in cases where $\|\theta\|$ is large (e.g. scenarios 2 and 5). It also explains why segment-ECE can produce valid correlation estimates in scenario 4, where standard ECE fails. Nevertheless, despite segmenting the data first, the power of the test is still greatly diminished.

These simulations demonstrate that ECE is robust to a wide variety of non-stationary settings, but is improved by applying segmentation first. The simulations indicated that Pearson correlation has no control over type I error, but ECE does, at the cost of some power. These findings are consistent with the asymptotic theory established in Section 4, where we show that $\hat{\rho} \xrightarrow{d} N(\rho, \sigma_\rho^2)$. Next, we will demonstrate ECE's utility on real data sets and compare them against alternative methods.

Table 4: Correlation matrix using labor productivity data

	DUR	NDUR	BUS	NFBUS	NFC
DUR	1.00	0.43	0.36	0.35	0.40
NDUR	0.43	1.00	0.37	0.36	0.15
BUS	0.36	0.37	1.00	0.98	0.32
NFBUS	0.35	0.36	0.98	1.00	0.33
NFC	0.40	0.15	0.32	0.33	1.00

(a) Pearson

	DUR	NDUR	BUS	NFBUS	NFC
DUR	1.00	0.23	0.27	0.27	0.25
NDUR	0.23	1.00	0.32	0.32	0.10
BUS	0.27	0.32	1.00	0.98	0.32
NFBUS	0.27	0.32	0.98	1.00	0.33
NFC	0.25	0.10	0.32	0.33	1.00

(b) Segmentation

	DUR	NDUR	BUS	NFBUS	NFC
DUR	1.00	0.33	0.32	0.32	0.36
NDUR	0.33	1.00	0.32	0.32	0.09
BUS	0.32	0.32	1.00	0.98	0.28
NFBUS	0.32	0.32	0.98	1.00	0.30
NFC	0.36	0.09	0.28	0.30	1.00

(c) Smoothing

	DUR	NDUR	BUS	NFBUS	NFC
DUR	1.00	0.06	-0.21	-0.16	0.08
NDUR	0.06	1.00	0.47	0.48	-0.06
BUS	-0.21	0.47	1.00	0.97	0.04
NFBUS	-0.16	0.48	0.97	1.00	0.05
NFC	0.08	-0.06	0.04	0.05	1.00

(d) ECE

5.2 Labor Productivity Data

Building on the analysis of labor productivity by Hao et al. [2022], we apply our method to quarterly data from the U.S. Bureau of Labor Statistics [1]. Like them, we limit our analysis to the time period between 1987Q1 and 2019Q4, and focus on five sectors: durable manufacturing (`DUR`), nondurable manufacturing (`NDUR`), business (`BUS`), non-farm business (`NFBUS`), and non-financial corporations (`NFC`). These time series are interesting because they possess a hierarchical structure: `BUS` mostly contains `NFBUS`, which can be divided further into `NFC`, `DUR`, `NDUR`, and other sub-sectors. This, in turn, implies a correlation within the hierarchy. On top of that, this data exhibits both stationarity and non-stationarity, making it an interesting test case for our estimator.

We will use two mean-removal techniques to analyze the data: segmentation and *smoothing*. The segmentation method, as previously discussed, estimates the average between change points, subtracts the estimated mean to make the data stationary, and then applies Pearson correlation. The smoothing approach is similar, but estimates the mean using non-parametric smoothing, subtracts the estimated mean, and then applies Pearson correlation on the residuals [Nilsen et al., 2012]. To implement this, we use simple exponential smoothing with the `ses()` function from the `forecast` package in `R`. We use the default parameters for simplicity. Both segmentation and smoothing approaches have the drawback of requiring tuning parameters, and the resulting estimates can be sensitive to their specification. We compare these mean-removal techniques against ECE on labor productivity data.

[1] Data is publicly available at `https://www.bls.gov/productivity/tables/`

Table 5: p-values for correlation test

	DUR	NDUR	BUS	NFBUS	NFC
DUR	.	0.00	0.00	0.00	0.00
NDUR	0.00	.	0.00	0.00	0.09
BUS	0.00	0.00	.	0.00	0.00
NFBUS	0.00	0.00	0.00	.	0.00
NFC	0.00	0.09	0.00	0.00	.

(a) Pearson

	DUR	NDUR	BUS	NFBUS	NFC
DUR	.	0.01	0.00	0.00	0.00
NDUR	0.01	.	0.00	0.00	0.30
BUS	0.00	0.00	.	0.00	0.00
NFBUS	0.00	0.00	0.00	.	0.00
NFC	0.00	0.30	0.00	0.00	.

(b) Segmentation

	DUR	NDUR	BUS	NFBUS	NFC
DUR	.	0.00	0.00	0.00	0.00
NDUR	0.00	.	0.00	0.00	0.30
BUS	0.00	0.00	.	0.00	0.00
NFBUS	0.00	0.00	0.00	.	0.00
NFC	0.00	0.30	0.00	0.00	.

(c) Smoothing

	DUR	NDUR	BUS	NFBUS	NFC
DUR	.	0.37	0.13	0.19	0.34
NDUR	0.37	.	0.00	0.00	0.36
BUS	0.13	0.00	.	0.00	0.39
NFBUS	0.19	0.00	0.00	.	0.38
NFC	0.34	0.36	0.39	0.38	.

(d) ECE

Table 4 reports the estimates of each of the aforementioned estimators. As expected, the naive Pearson method produces correlation estimates, likely mistaking mean shifts for variation. The mean-removal strategies correct for this bias and yield smaller correlation estimates. Still, the corresponding p-values are nearly identical to Pearson's (Table 5), limiting their ability to distinguish meaningful relationships. In contrast, ECE paints a much clearer picture: for pairs with significant p-values, such as `NDUR-BUS` and `NDUR-NFBUS`, ECE produces estimates comparable to the mean-removal methods; but for non-significant pairs, ECE further deflates the correlations. This pattern suggests that ECE may account for subtle, undetected structural breaks. As a result, ECE highlights only a small subset of interactions as significant, avoiding the tendency of other methods to label nearly all relationships as correlated.

5.3 U.S. Employment

The Federal Reserve maintains records of various macroeconomic indicators across time. McCracken and Ng [2015] have curated FRED-MD, a monthly dataset, and regularly maintain it [2]. Their working paper provides recommended transformations to make each time series stationary, and the `fbi` R package provides a user-friendly approach to load and transform the data.

We analyze a selection of macroeconomic indicators from the FRED-MD database between January 1959 and January 2024. There are 119 macroeconomic indicators across 780

[2]Data can be found at `https://research.stlouisfed.org/econ/mccracken/fred-databases`

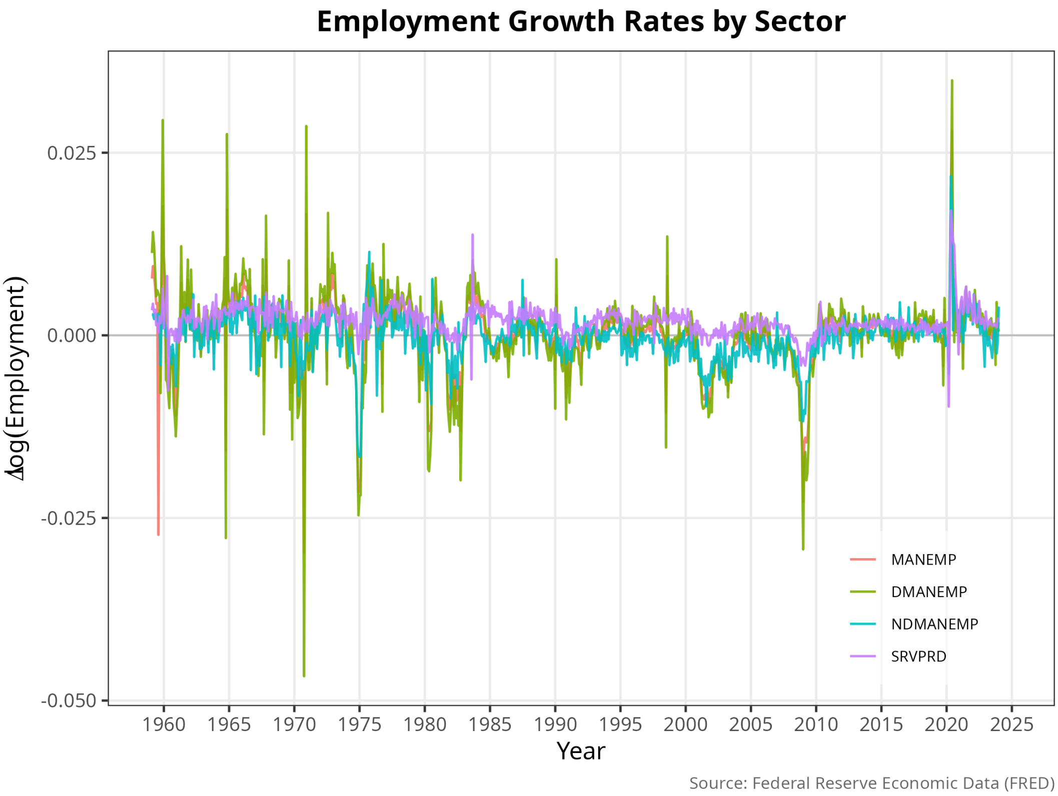

Employment Growth Rates by Sector
0.025
0.000
-0.025
-0.050
Δlog(Employment)
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
MANEMP
DMANEMP
NDMANEMP
SRVPRD
Source: Federal Reserve Economic Data (FRED)

Figure 5

months, 42 of which contain a missing value in at least one of the indicators, which will be imputed with linear interpolation.

All macroeconomic indicators investigated in this report undergo a first log-difference transformation ($\Delta \log(x_t)$) to stabilize variance and remove trends. However, as shown in Figure 5, residual non-stationarity persists, indicating shocks, which can be approximated by a piecewise constant model. There is a significant reduction in employment after the 2008 recession, and a sharp increase in employment after the COVID pandemic.

We analyze U.S. employment in four sectors: all manufacturing (`MANEMP`), durable manufacturing (`DMANEMP`), non-durable manufacturing (`NDMANEMP`), and services (`SRVPRD`). These sectors are chosen to highlight our estimator's ability to selectively identify statistical correlations between related sectors while correctly denying statistical correlations between unrelated sectors. `MANEMP` is correlated with `DMANEMP` and `NDMANEMP` because `MANEMP = DMANEMP + NDMANEMP`. Generally, the US economy can be divided into goods and services. Goods are tangible products (e.g. cars, textile, electronics), whereas services are intangible activities (e.g. banking, consulting, healthcare). Manufacturing can be further broken down into durables and non-durables.

Intuitively, the forces that influence manufacturing differ from the forces that influence the service industry. At a minimum, we expect the estimated correlation within the manufacturing sectors to be greater than the correlation between the manufacturing and services sectors.

As expected, `MANEMP` is significantly correlated with both `DMANEMP` and `NDMANEMP`, but not `SRVPRD`. Interestingly enough, despite both being types of manufacturing, `DMANEMP` and `NDMANEMP` are not correlated. We find that `SRVPRD` does not have any significant cross-sectional dependence with the manufacturing sectors.

Like in the previous real data examples, we will compare ECE against segmentation. Once again, we observe that segmentation deems each macroeconomic indicator to be cross-sectionally correlated. There are strong correlations between each sector because segmentation fails to adequately fit the data. Therefore, the obvious non-stationarities get interpreted as genuine covariation.

	MANEMP	DMANEMP	NDMANEMP	SRVPRD
MANEMP	.	0.95	0.75	0.56
DMANEMP	0.95	.	0.62	0.51
NDMANEMP	0.75	0.62	.	0.57
SRVPRD	0.56	0.51	0.57	.

Table 6: Pearson Correlation

	MANEMP	DMANEMP	DMANEMP	SRVPRD
MANEMP	.	0.87	0.28	0.07
DMANEMP	0.87	.	0.02	0.02
NDMANEMP	0.28	0.02	.	0.13
SRVPRD	0.07	0.02	0.13	.

Table 7: ECE Correlation

	MANEMP	DMANEMP	NDMANEMP	SRVPRD
MANEMP	.	0.00	0.00	0.00
DMANEMP	0.00	.	0.00	0.00
NDMANEMP	0.00	0.00	.	0.00
SRVPRD	0.00	0.00	0.00	.

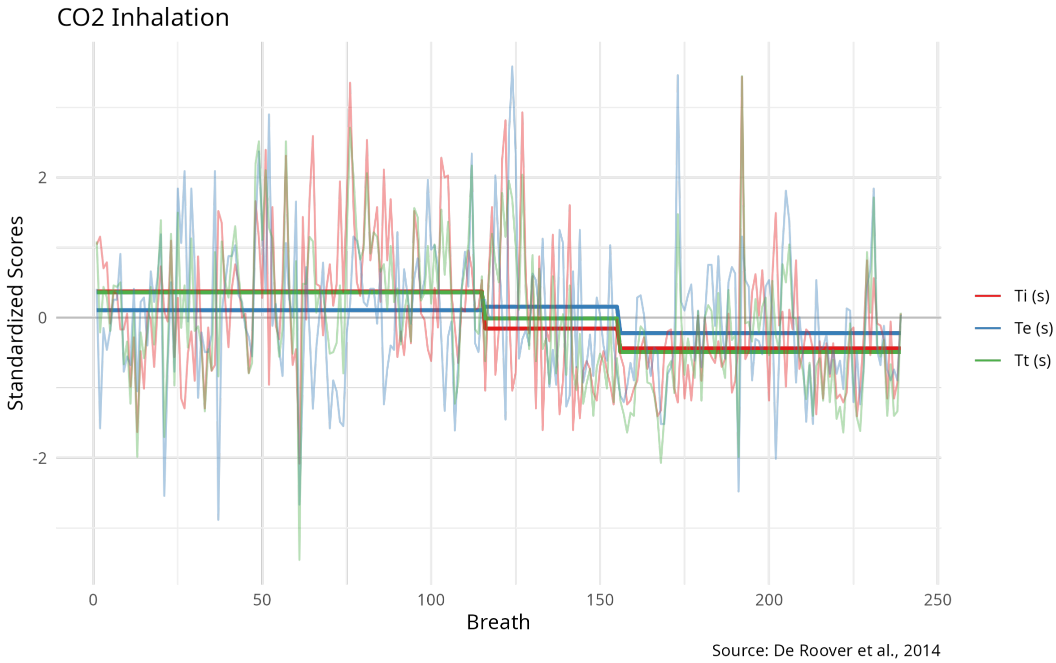
Table 8: Pearson p-Values

	MANEMP	DMANEMP	NDMANEMP	SRVPRD
MANEMP	.	0.00	0.00	0.30
DMANEMP	0.00	.	0.82	0.78
NDMANEMP	0.00	0.82	.	0.06
SRVPRD	0.30	0.78	0.06	.

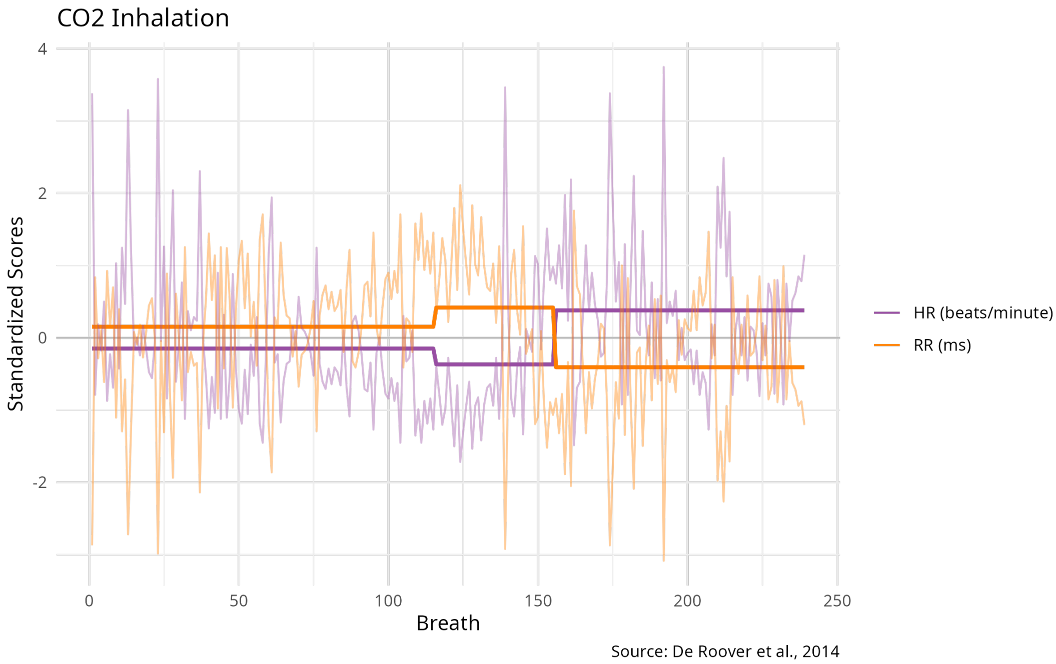
Table 9: ECE p-Values

5.4 CO_2 Inhalation

Finally, we compare ECE against Pearson correlation, where controlled experimental conditions determine the location of change points. By knowing the change points, we can compare the segmentation method, which explicitly adjusts for mean shifts against ECE, which only implicitly adjusts for the mean shifts. Since the change points are known, they



(a)



(b)

Figure 6: Time Series

provide a natural way to remove means and make the data stationary. Ideally, both methods produce similar results.

De Roover et al. [2014] collect physiological data from an individual over time. The individual wears a gas mask that allows the experimenters to control whether the individual breathes regular air of a CO_2 mixture. He is asked to breathe normally while the air mixture is discreetly changed without informing the subject. The experiment begins with a resting baseline (11 min.), followed by a CO_2 inhalation period (5 min.), and finally, a recovery period (6 min.). With each breath, respiratory data such as the length of inhalation (`Ti`), length of exhalation (`Te`), and the total length of the respiratory cycle (`Tt = Ti + Te`) are recorded. They also measure cardiac data such as heart rate (`HR`) and the cardiac interbeat interval (`RR`) – the time between peaks on an electrocardiogram. Together, this forms a time series where each index represents a breath.

In total, they collect 290 data points corresponding to each breath. However, the subject's movement from coughs and hiccups produces measurement error, so they removed an additional 51 observations, resulting in a total of 239 data points. Finally, they normalize each time series to have zero mean and unit variance. We access this data directly from the `kcpRS` R package [Cabrieto et al., 2021].

This dataset has the added benefit of having known correlation directions. For example, we know `HR` and `RR` are inversely correlated, because a higher heart rate corresponds to a shorter distance between peaks on an electrocardiogram. Moreover, since `Tt` is the sum of `Ti` and `Te` prior to normalization, we can expect `Tt-Ti` and `Tt-Te` to be positively correlated.

Figure 6 illustrates the time series for respiratory times and cardiac measurements. Judging from the observed data, the heart rate (`HR`) increases during the CO_2 inhalation phase, and drops in recovery, whereas the interbeat interval (`RR`) inverts this behavior. Alternatively, the respiratory times decrease during the CO_2 inhalation phase and then recover in the final phase. It is established in the literature that subjects who inhale CO_2 respond with increased heart rate and shortened breath [Bailey et al., 2005]. Despite this, the average respiratory times of each phase tell a different story; average respiratory times decrease in each phase of the experiment. This is because the effects of inhaling CO_2 or regular air do not immediately correspond to the change in experimental conditions. It is well known that there is a lag between when the air mixture changes and when the subject's physiology re-

	Ti	Te	Tt	HR	RR
Ti	1.00	0.07	0.73	0.18	-0.17
Te	0.07	1.00	0.73	0.19	-0.22
Tt	0.73	0.73	1.00	0.25	-0.26
HR	0.18	0.19	0.25	1.00	-0.99
RR	-0.17	-0.22	-0.26	-0.99	1.00

(a) ECE Correlation

	Ti	Te	Tt	HR	RR
Ti	1.00	-0.13	0.68	-0.12	0.15
Te	-0.13	1.00	0.64	-0.08	0.09
Tt	0.68	0.64	1.00	-0.15	0.18
HR	-0.12	-0.08	-0.15	1.00	-0.99
RR	0.15	0.09	0.18	-0.99	1.00

(b) Segment (Oracle) Correlation

Table 10: Correlation

	Ti	Te	Tt	HR	RR
Ti	.	0.62	**0.00**	0.14	0.18
Te	0.62	.	**0.00**	0.13	0.08
Tt	**0.00**	**0.00**	.	**0.02**	**0.02**
HR	0.14	0.13	**0.02**	.	**0.00**
RR	0.18	0.08	**0.02**	**0.00**	.

(a) ECE p-Value

	Ti	Te	Tt	HR	RR
Ti	.	0.05	**0.00**	0.06	**0.02**
Te	0.05	.	**0.00**	0.24	0.15
Tt	**0.00**	**0.00**	.	**0.02**	**0.00**
HR	0.06	0.24	**0.02**	.	**0.00**
RR	**0.02**	0.15	**0.00**	**0.00**	.

(b) Segment (Oracle) p-Value

Table 11: p-Values

sponds. On top of this, there is even some variation within the same experimental condition. All of this shows that despite knowing when the gas changes, it may still be insufficient to capture the non-stationarity of the process.

Tables 10a-10b show the estimated correlations and Tables 11a-11b show the corresponding p-values. Both methods correctly identify that `Ti` and `Te` are pairwise-correlated with `Tt`. They are also able to identify the expected negative correlation between `HR` and `RR`.

This application is one where the literature supports the notion that CO_2 inhalation produces shortened breath and increased heart rate. Naively applying Pearson correlation increases spurious correlations. However, by knowing the experimental conditions, we can remove non-stationarity via segmentation. Unfortunately, as we have shown, even if the change points are known, lags in physiological processes may limit their effectiveness at removing mean shifts. We demonstrate that ECE can correctly identify correlated time series while automatically adjusting for mean shifts.

6 Discussion

Non-stationarity is a ubiquitous feature of modern data. Econometrics data is collected over decades and must contend with structural breaks stemming from policy shifts, market crashes, and global disruptions. Naively applying classical methods such as Pearson correlation without accounting for these structural breaks can produce false conclusions, making them wholly inadequate for analysis of complex data. Instead, it motivates the development of techniques explicitly designed for data with structural breaks.

In this report, we expand on the equivariant variance estimator (EVE) [Hao et al., 2022], and propose the equivariant covariance estimator (ECE). Assuming a data sequence follows a piecewise constant mean structure, EVE was limited in scope to estimating the sequence's variance. ECE extends this scope to estimating the covariance matrix for bivariate data sequences. Our method enjoys many properties that make it appealing for estimating contemporaneous correlation. Not only do we prove that it is unbiased and equivariant, but among all quadratic estimators, it is the *unique* unbiased equivariant estimator. Because practitioners often interpret dependence through correlation, we further adapt ECE to estimate correlation and produce a corresponding hypothesis testing procedure.

To evaluate its finite sample properties and robustness to model misspecification, we compare ECE against some competitors in a wide range of simulation settings. Competing methods fail to consistently control type I error, casting doubt on their utility on real world non-stationary data. They fail in part because they struggle to accurately identify change points, especially if they are small and frequent. However, our simulations indicate that even if the change points are known, mean-removal strategies may still fail at controlling type I error. Moreover, type I error control may not improve with increasing sample sizes. In fact, increased sample sizes may even undermine its ability to identify change points. In contrast, our method is robust. It can control type I error, even if the underlying mean-structure is unspecified. For smaller sample sizes, it exhibits more false positives than expected under the null, but this vanishes with increasing sample size. ECE is also modestly powered. While it has less power than Pearson correlation without change points, it is a small price to pay for the ability to accurately estimate correlation between sequences with mean shifts.

Applications to labor productivity, US employment, and experimental physiological data highlight ECE's interpretive and practical advantages. Competing techniques tend to be overly sensitive, detecting significant correlations across most variables, obfuscating the interpretive significance of any of the results. ECE, by contrast, produces more selective determinations of significance while maintaining the ability to detect obviously correlated variables, such as total manufacturing employment with its constituent durable and non-durable sub-sectors, and total respiratory time with its constituent inhalation and exhalation times.

While we have shown promising preliminary results for our method, we have made several simplifying assumptions. We assume Gaussian errors, neglecting the skew and heavy tails that are often present in real-world data. In addition to simplifying the model in this analysis, we also made simplifications to our hypothesis testing procedure. We limited our analysis to the null, which allows us to make determinations on statistical significance of the contemporaneous correlation between variables, but it is insufficient to produce confidence intervals, which requires extending the asymptotic analysis for $\rho \neq 0$. Finally, our analysis is restricted to bivariate data. There are no guarantees filling a multivariate covariance matrix with the pairwise covariances will produce a positive semi-definite matrix.

All in all, our findings demonstrate that ECE is a principled and robust method for estimating cross-sectional correlation in the presence of mean shifts, with applicability to a broad array of non-stationary time-series data.

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Appendix

Proofs for Section 3

Proof of Proposition 1.

$$
\begin{aligned}
\mathbb{E}\left(T_k(X,Y)\right) &= \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[(X_i - X_{i+k})(Y_i - Y_{i+k})\right] \\
&= \frac{1}{n}\sum_{i=1}^{n}\Big[(\theta_{x,i} - \theta_{x,i+k})(\theta_{y,i} - \theta_{y,i+k}) - \\
&\qquad\quad (\theta_{x,i} - \theta_{x,i+k})\mathbb{E}(\varepsilon_{y,i} - \varepsilon_{y,i+k}) - \\
&\qquad\quad (\theta_{y,i} - \theta_{y,i+k})\mathbb{E}(\varepsilon_{x,i} - \varepsilon_{x,i+k}) + \\
&\qquad\quad \mathbb{E}\left((\varepsilon_{x,i} - \varepsilon_{x,i+k})(\varepsilon_{y,i} - \varepsilon_{y,i+k})\right)\Big] \\
&= \frac{1}{n}\Big[\sum_{i=1}^{n}(\theta_{x,i} - \theta_{x,i+k})(\theta_{y,i} - \theta_{y,i+k}) + \sum_{i=1}^{n}\mathbb{E}\left((\varepsilon_{x,i} - \varepsilon_{x,i+k})(\varepsilon_{y,i} - \varepsilon_{y,i+k})\right)\Big].
\end{aligned}
$$

Now, we find that

$$
\begin{aligned}
&\mathbb{E}\left((\varepsilon_{x,i} - \varepsilon_{x,i+k})(\varepsilon_{y,i} - \varepsilon_{y,i+k})\right) \\
&= \mathbb{E}\left(\varepsilon_{x,i}\varepsilon_{y,i} - \varepsilon_{x,i}\varepsilon_{y,i+k} - \varepsilon_{x,i+k}\varepsilon_{y,i} + \varepsilon_{x,i+k}\varepsilon_{y,i+k}\right) \\
&= 2\sigma_{xy},
\end{aligned}
$$

because $\mathbb{E}\left(\varepsilon_{x,i}\varepsilon_{y,i+k}\right) = \mathbb{E}\left(\varepsilon_{x,i+k}\varepsilon_{y,i}\right) = 0$.

We also have

$$\begin{aligned}
&\sum_{i=1}^{n}(\theta_{x,i} - \theta_{x,i+k})(\theta_{y,i} - \theta_{y,i+k}) \\
&= T_k(\theta_x, \theta_y) \\
&= kT_1(\theta_x, \theta_y).
\end{aligned}$$

Therefore,

$$\mathbb{E}\left(T_k(X,Y)\right) = kT_1(\theta_x, \theta_y) + 2\sigma_{xy}.$$

It follows easily that

$$\begin{aligned}
\mathbb{E}\left(T_k(X)\right) &= kT_1(\theta_x) + 2\sigma_x^2 \\
\mathbb{E}\left(T_k(Y)\right) &= kT_1(\theta_y) + 2\sigma_y^2.
\end{aligned}$$

Proof of Theorem 2. We introduce some new notation to facilitate this proof. First, let the collection of break points that do not occur consecutively be

$$\mathcal{T}_J = \left\{\boldsymbol{\tau} = (\tau_1, ..., \tau_J) : \tau_{i+1} - \tau_i \geq 2, i = 1, ..., J\right\}. \tag{17}$$

Define the model class Θ as the collection of bivariate, piecewise-constant functions, with structural breaks at $\boldsymbol{\tau} \in \mathcal{T}_J$:

$$\Theta = \left\{\theta = \begin{bmatrix}\theta_x & \theta_y\end{bmatrix} \in \mathbb{R}^{n\times 2} : \theta_i = \theta_{i+1}, \quad \forall (i+1) \notin \boldsymbol{\tau}\right\}. \tag{18}$$

We also want to express any piecewise-constant function as a composition of its constant segments. We achieve this by defining a support for each segment, and a corresponding value representing the value of θ at that support.

To begin, let $[n] = \{1, 2, \ldots, n\}$ denote the circular index set as described in Section 2.

We define a segment $\Lambda = [a, b] \subseteq [n]$ as a set of contiguous indices (wrapping around if necessary). Then, we define $\mathcal{I}$ to be the collection of all contiguous segments on a circular index set:

$$\mathcal{I} = \{\Lambda = [a, b] \subseteq [n] : 2 \leq |\Lambda| \leq n/2\}.$$

We also want to express a piecewise-constant function in terms of its constituent segments. Consider $\theta = [\theta_x \quad \theta_y] \in \Theta$ with M segments, we can decompose θ_x and θ_y into

$$\theta_x = \sum_{m=1}^{M} \mu_{m,x} \, \mathbb{1}_{\Lambda_m},$$
$$\theta_y = \sum_{m=1}^{M} \mu_{m,y} \, \mathbb{1}_{\Lambda_m},$$

where $\mu_{x,m}$ and $\mu_{y,m}$ denotes the constant value of θ_x and θ_y on its mth segment, and the support of the mth segment can be represented by the indicator function $\mathbb{1}_{\Lambda_m}$.

Now, we are ready to prove this theorem. We can write quadratic estimators as $X^T AY$ for $n \times n$ matrix A. Expanding the expectation, we find

$$\begin{aligned}
\mathbb{E}\left[X^T AY\right] &= \mathbb{E}\left[(\theta_x + \varepsilon_x)^T A(\theta_y + \varepsilon_y)\right] \\
&= \theta_x^T A\theta_y + \mathbb{E}\left[\varepsilon_x^T A\varepsilon_y\right] \\
&= \theta_x^T A\theta_y + \sigma_{xy}\mathrm{tr}(A).
\end{aligned}$$

where the last line follows since ε_x and ε_y are not auto-correlated, so only the diagonal elements of $\mathbb{E}\left(\varepsilon_x^T A\varepsilon_y\right)$ are non-zero. Therefore, $X^T AY$ is unbiased if and only if

$$\theta_x^T A\theta_y = 0, \quad \forall \theta_x, \theta_y \in \Theta,$$

and $\mathrm{tr}(A) = 1$.

We transform the problem by decomposing the mean vectors into $\theta_x = \sum_{m_x=1}^{M} \mu_{m_x} \mathbb{1}_{\Lambda_m}$, and $\theta_y = \sum_{m_y=1}^{M} \mu_{m_y} \mathbb{1}_{\Lambda_m}$, where $\Lambda_m \in \mathcal{I}$. Then,

$$\theta_x^T A \theta_y = \sum_{m_x=1}^{M} \sum_{m_y=1}^{M} \mu_{m_x} \mu_{m_y} \mathbb{1}_{\Lambda_m}^T A \mathbb{1}_{\Lambda_m} = 0$$

if and only if

$$\mathbb{1}_{\Lambda_m}^T A \mathbb{1}_{\Lambda_m} = 0, \quad \forall \Lambda_m \in \mathcal{I}. \tag{19}$$

By hypothesis, $X^T AY$ is equivariant, so A must be a circulant matrix, defined as

$$\bar{A} = \begin{bmatrix} a_0 & a_{n-1} & a_{n-2} & \dots & a_3 & a_2 & a_1 \\ a_1 & a_0 & a_{n-1} & \dots & a_4 & a_3 & a_2 \\ a_2 & a_1 & a_0 & \dots & a_5 & a_4 & a_3 \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots \\ a_{n-3} & a_{n-4} & a_{n-5} & \dots & a_0 & a_{n-1} & a_{n-2} \\ a_{n-2} & a_{n-3} & a_{n-4} & \dots & a_1 & a_0 & a_{n-1} \\ a_{n-1} & a_{n-2} & a_{n-3} & \dots & a_2 & a_1 & a_0 \end{bmatrix}. \tag{20}$$

In other words, to get an unbiased estimate for σ_{xy}, we must find A where each *principal block sum* is 0. Consequently, the formula for the principal block sum is

$$g(l) = la_0 + \sum_{i=1}^{l} 2(l-i)a_i, \quad l = 1, 2, ..., n. \tag{21}$$

Without loss of generality, we may assume that A is symmetric since the principal block sums are invariant with respect to symmetry for circulant A.

We denote symmetric A as

$$A = \begin{bmatrix} a_0 & a_1 & a_2 & \dots & a_3 & a_2 & a_1 \\ a_1 & a_0 & a_1 & \dots & a_4 & a_3 & a_2 \\ a_2 & a_1 & a_0 & \dots & a_5 & a_4 & a_3 \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots \\ a_3 & a_4 & a_5 & \dots & a_0 & a_1 & a_2 \\ a_2 & a_3 & a_4 & \dots & a_1 & a_0 & a_1 \\ a_1 & a_2 & a_3 & \dots & a_2 & a_1 & a_0 \end{bmatrix}. \tag{22}$$

Consequently, the formula for the principal block sums for a symmetric A is

$$g(l) = la_0 + \sum_{i=1}^{\min\{\lfloor n/2 \rfloor, l\}} 2(l-i)a_i + \sum_{i=\lfloor n/2 \rfloor+1}^{l} 2(l-i)a_{n-i}. \tag{23}$$

To find A, we ensure that

$$g(l) = 0, \text{ for } l = 2, 3, ..., \lceil n/2 \rceil. \tag{24}$$

Consider $l = 2$:

$$g(2) = 2a_0 + 2a_1 = 0$$
$$\implies a_1 = -a_0$$

For $l = 3$,

$$g(3) = 3a_0 + 4a_1 + 2a_2 = 0$$
$$\implies a_2 = a_0/2$$

We show by induction that $a_{l-1} = 0$ satisfies $g(l) = 0$ for $4 \leq l \leq n-2$. Consider the base case:

$$\begin{aligned}
g(4) &= 4a_0 + 6a_1 + 4a_2 + 2a_3 \\
&= 4a_0 - 6a_0 + 2a_0 + 2a_3 \\
\implies a_3 &= 0
\end{aligned}$$

Suppose that for $g(l) = 0$, $a_{l-1} = 0$. Then, we see that $g(l+1), a_l = 0$.

$$\begin{aligned}
g(l+1) &= (l+1)a_0 + \sum_{i=1}^{l+1} 2(l+1-i)a_i \\
&= la_0 + \sum_{i=1}^{l} 2(l-i)a_i + a_0 + \sum_{i=1}^{l} 2a_i \\
&= g(l) + a_0 + \sum_{i=1}^{l} 2a_i \\
&= a_0 + 2a_1 + 2a_2 + \sum_{i=3}^{l} 2a_i \\
&= 0 + \sum_{i=3}^{l-1} 2a_i + 2a_l \\
\implies a_l &= 0.
\end{aligned}$$

Finally, we validate that for $\lceil n/2 \rceil < l \leq n-2$, $g(l) = 0$.

$$\begin{aligned}
g(l) &= la_0 + \sum_{i=1}^{\lfloor n/2 \rfloor} 2(l-i)a_i + \sum_{i=\lfloor n/2 \rfloor + 1}^{l} 2(l-i)a_{n-i} \\
&= la_0 + 2(l-1)a_1 + 2(l-2)a_2 + 0 \\
&= la_0 - 2(l-1)a_0 + (l-2)a_0 \\
&= 0
\end{aligned}$$

Lastly, for $l = n$,

$$
\begin{aligned}
g(n) &= na_0 + \sum_{i=1}^{\lfloor n/2 \rfloor} 2(n-i)a_i + \sum_{i=\lfloor n/2 \rfloor+1}^{n} 2(n-i)a_{n-i} \\
&= na_0 + 2(n-1)a_1 + 2(n-2)a_2 + 2(n-(n-2)) + 2(n-(n-1)) \\
&= na_0 + 2na_1 - 2a_1 + 2na_2 - 4a_2 + 4a_2 + 2a_1 \\
&= na_0 - 2na_0 + 2a_0 + na_0 - 2a_0 + 2a_0 - 2a_0 \\
&= 0
\end{aligned}
$$

Therefore, we find that the A that satisfies $\mathbb{1}_{\Lambda_m}^T A \mathbb{1}_{\Lambda_m} = 0, \forall \Lambda_m \in \mathcal{I}$ has values $a = (1, -1, 1/2, 0, ..., 0, 1/2, 1)^T$. Rearranging terms, $X^T AY$ is the same as ECE as described in Section 3.

Proofs for Section 4

Proof of Lemma 2. We must show that $T_k(X, Y)$ is jointly normal asymptotically. We begin by decomposing it as follows:

$$
\begin{aligned}
T_k(X,Y) &= \sum_{i=1}^{n} (X_i - X_{i+k})(Y_i - Y_{i+k}) \\
&= \sum_{i=1}^{n} (\theta_{x,i} - \theta_{x,i+k} + \varepsilon_{x,i} - \varepsilon_{x,i+k})(\theta_{y,i} - \theta_{y,i+k} + \varepsilon_{y,i} - \varepsilon_{y,i+k}) \\
&= \sum_{i=1}^{n} (\theta_{x,i} - \theta_{x,i+k})(\theta_{y,i} - \theta_{y,i+k}) + \sum_{i=1}^{n} (\varepsilon_{x,i} - \varepsilon_{x,i+k})(\theta_{y,i} - \theta_{y,i+k}) + \\
&\quad \sum_{i=1}^{n} (\theta_{x,i} - \theta_{x,i+k})(\varepsilon_{y,i} - \varepsilon_{y,i+k}) + \sum_{i=1}^{n} (\varepsilon_{x,i} - \varepsilon_{x,i+k})(\varepsilon_{y,i} - \varepsilon_{y,i+k}).
\end{aligned}
$$

Now, we will show that each of these terms is constant or asymptotically normal, so the sum of all the terms is asymptotically normal. Clearly, $\sum_{i=1}^{n} (\theta_{x,i} - \theta_{x,i+k})(\theta_{y,i} - \theta_{y,i+k})$ is constant. Next, we see that

$$\begin{aligned}&\sum_{i=1}^{n}(\varepsilon_{x,i}-\varepsilon_{x,i+k})(\varepsilon_{y,i}-\varepsilon_{y,i+k})\\&=\sum_{i=1}^{n}(\varepsilon_{x,i}\varepsilon_{y,i}-\varepsilon_{x,i+k}\varepsilon_{y,i}-\varepsilon_{x,i}\varepsilon_{y,i+k}+\varepsilon_{x,i+k}\varepsilon_{y,i+k}),\end{aligned}$$

so it is a sum of weakly dependent and identically distributed random variables, thus asymptotically normal. Finally, we decompose $\sum_{i=1}^{n}(\varepsilon_{x,i}-\varepsilon_{x,i+k})(\theta_{y,i}-\theta_{y,i+k})$ so that

$$\begin{aligned}&\sum_{i=1}^{n}(\varepsilon_{x,i}-\varepsilon_{x,i+k})(\theta_{y,i}-\theta_{y,i+k})\\&=\sum_{i=1}^{n}\varepsilon_{x,i}(\theta_{y,i}-\theta_{y,i+k})-\sum_{i=1}^{n}\varepsilon_{x,i+k}(\theta_{y,i}-\theta_{y,i+k}).\end{aligned}$$

Since we enforce the Lindeberg condition by hypothesis, we get the sum of two random variables by the central limit theorem. Therefore, $\sum_{i=1}^{n}(\varepsilon_{x,i}-\varepsilon_{x,i+k})(\theta_{y,i}-\theta_{y,i+k})$ and $\sum_{i=1}^{n}(\theta_{x,i}-\theta_{x,i+k})(\varepsilon_{y,i}-\varepsilon_{y,i+k})$ can be represented asymptotically as the sum of normal random variables. The argument for the asymptotic normality of $\sum_{i=1}^{n}(\theta_{x,i}-\theta_{x,i+k})(\varepsilon_{y,i}-\varepsilon_{y,i+k})$ follow similarly.

Altogether, we have shown that $T_k(X,Y)$ is normally distributed. Thus, since we can represent V in terms of $T_k(X,Y)$:

$$V=\begin{pmatrix}T_1(X) & T_2(X) & T_1(Y) & T_2(Y) & T_1(X,Y) & T_2(X,Y)\end{pmatrix}^T,$$

and we have shown that each element is asymptotically normal, then, any linear combination of them is normally distributed. That is, for any $a\in\mathbb{R}^6$, it follows that a^TV is also normally distributed. By the Cramér-Wold device, V is asymptotically normal.

Proof of Proposition 2a $(\mathrm{Var}(T_k(X,Y)))$.

We begin by decomposing the variance of $T_k(X,Y)$ into the sum of diagonal and off-diagonal components:

$$
\begin{aligned}
\mathrm{Var}(T_k(X,Y)) &= \mathrm{Var}\left(\sum_{i=1}^{n}(X_i - X_{i+k})(Y_i - Y_{i+k})\right)\\
&= \sum_{i=1}^{n} \mathrm{Var}\left((X_i - X_{i+k})(Y_i - Y_{i+k})\right)\\
&\quad + \sum_{i\neq i'} \mathrm{Cov}\left((X_i - X_{i+k})(Y_i - Y_{i+k}), (X_{i'} - X_{i'+k})(Y_{i'} - Y_{i'+k})\right).
\end{aligned}
$$

We expand this expression by applying Lemmas 3 and 4, which evaluate $\mathrm{Var}\left((X_i - X_{i+k})(Y_i - Y_{i+k})\right)$ and $\mathrm{Cov}\left((X_i - X_{i+k})(Y_i - Y_{i+k}), (X_{i'} - X_{i'+k})(Y_{i'} - Y_{i'+k})\right)$:

$$
\begin{aligned}
&\mathrm{Var}(T_k(X,Y))\\
=&\sum_{i=1}^{n} \mathrm{Var}\left((X_i - X_{i+k})(Y_i - Y_{i+k})\right)\\
&+ \sum_{i\neq i'} \mathrm{Cov}\left((X_i - X_{i+k})(Y_i - Y_{i+k}), (X_{i'} - X_{i'+k})(Y_{i'} - Y_{i'+k})\right)\\
=&\sum_{i=1}^{n}\left[2(\theta_{x,i} - \theta_{x,i+k})^2 + 2(\theta_{y,i} - \theta_{y,i+k})^2 + 2\kappa_{22} + 2 + 4\rho(\theta_{x,i} - \theta_{x,i+k})(\theta_{y,i} - \theta_{y,i+k})\right]\\
&+ 2\sum_{i=1}^{n} \mathrm{Cov}\left((X_i - X_{i+k})(Y_i - Y_{i+k}), (X_{i+k} - X_{i+2k})(Y_{i+k} - Y_{i+2k})\right)\\
=&2n(\kappa_{22}+1) + 2\sum_{i=1}^{n}(\theta_{x,i} - \theta_{x,i+k})^2 + 2\sum_{i=1}^{n}(\theta_{y,i} - \theta_{y,i+k})^2 + 4\rho\sum_{i=1}^{n}(\theta_{x,i} - \theta_{x,i+k})(\theta_{y,i} - \theta_{y,i+k})\\
&+ 2\sum_{i=1}^{n}\Big[(\kappa_{22}-\rho^2) - (\theta_{x,i} - \theta_{x,i+k})(\theta_{x,i+k} - \theta_{x,i+2k}) - (\theta_{y,i} - \theta_{y,i+k})(\theta_{y,i+k} - \theta_{y,i+2k})\\
&\quad - \rho(\theta_{x,i} - \theta_{x,i+k})(\theta_{y,i+k} - \theta_{y,i+2k}) - \rho(\theta_{y,i} - \theta_{y,i+k})(\theta_{x,i+k} - \theta_{x,i+2k})\\
&\quad - \kappa_{12}(\theta_{x,i} - 2\theta_{x,i+k} + \theta_{x,i+2k}) - \kappa_{21}(\theta_{y,i} - 2\theta_{y,i+k} + \theta_{y,i+2k})\Big]\\
=&2n(\kappa_{22}+1) + 2kT_1(\theta_x) + 2kT_1(\theta_y) + 4k\rho T_1(\theta_x,\theta_y) + 2n(\kappa_{22}-\rho^2)\\
=&2n(2\kappa_{22}+1-\rho^2) + 2kT_1(\theta_x) + 2kT_1(\theta_y) + 4k\rho T_1(\theta_x,\theta_y).
\end{aligned}
$$

This concludes the proof for $\mathrm{Var}(T_k(X,Y))$.

Proof of Proposition 2b $([\mathrm{Cov}(T_1(X,Y), T_2(X,Y))])$.

Now, we will show the proof of $\mathrm{Cov}(T_1(X,Y), T_2(X,Y))$.

$$\mathrm{Cov}(T_h(X,Y), T_k(X,Y)) = \sum_{i=1}^{n}\sum_{i'=1}^{n} \mathrm{Cov}\left((X_i - X_{i+k})(Y_i - Y_{i+k}), (X_{i'} - X_{i'+l})(Y_{i'} - Y_{i'+l})\right)$$

where the summands are nonzero only when $i = i'$, $i = i' + k$, $i + h = i'$, or $i + h = i' + k$. Then, by applying Lemma 4, we get

$$\begin{aligned}
\mathrm{Cov}(T_h(X,Y), T_k(X,Y)) =& \sum_{i=1}^{n} \mathrm{Cov}\left((X_i - X_{i+h})(Y_i - Y_{i+h}), (X_i - X_{i+k})(Y_i - Y_{i+k})\right) \\
&+ \sum_{i=1}^{n} \mathrm{Cov}\left((X_i - X_{i+h})(Y_i - Y_{i+h}), (X_{i-k} - X_i)(Y_{i-k} - Y_i)\right) \\
&+ \sum_{i=1}^{n} \mathrm{Cov}\left((X_i - X_{i+h})(Y_i - Y_{i+h}), (X_{i+h-k} - X_{i+h})(Y_{i+h-k} - Y_{i+h})\right) \\
&+ \sum_{i=1}^{n} \mathrm{Cov}\left((X_i - X_{i+h})(Y_i - Y_{i+h}), (X_{i+h} - X_{i+h+k})(Y_{i+h} - Y_{i+h+k})\right) \\
=& 4n(\kappa_{22} - \rho^2) + 2kT_1(\theta_x) + 2kT_1(\theta_y) + 4k\rho T_1(\theta_x, \theta_y).
\end{aligned}$$

Lemma 3 *Let $X_i = \theta_{x,i} + \varepsilon_{x,i}$ and $Y_i = \theta_{y,i} + \varepsilon_{y,i}$, where the error terms satisfy $\mathbb{E}[\varepsilon_{x,i}] = \mathbb{E}[\varepsilon_{y,i}] = 0$, $\mathrm{Var}(\varepsilon_{x,i}) = \mathrm{Var}(\varepsilon_{y,i}) = 1$, and $\mathrm{Cov}(\varepsilon_{x,i}, \varepsilon_{y,i}) = \rho$. Let $\kappa_{22} = \mathbb{E}[\varepsilon_{x,i}^2 \varepsilon_{y,i}^2]$. Then, for any lag k such that $1 \le k \le L/2$, we have*

$$\begin{aligned}
\mathrm{Var}\left[(X_i - X_{i+k})(Y_i - Y_{i+k})\right] =& 2(\theta_{x,i} - \theta_{x,i+k})^2 + 2(\theta_{y,i} - \theta_{y,i+k})^2 \\
&+ 2\kappa_{22} + 2 + 4\rho(\theta_{x,i} - \theta_{x,i+k})(\theta_{y,i} - \theta_{y,i+k}).
\end{aligned}$$

$$
\begin{aligned}
&\operatorname{Var}\left[(X_i - X_{i+k})(Y_i - Y_{i+k})\right] \\
&= \operatorname{Var}\left[(\theta_{x,i} - \theta_{x,i+k} + \varepsilon_{x,i} - \varepsilon_{x,i+k})(\theta_{y,i} - \theta_{y,i+k} + \varepsilon_{y,i} - \varepsilon_{y,i+k})\right] \\
&= \operatorname{Var}\Big[(\theta_{x,i} - \theta_{x,i+k})(\varepsilon_{y,i} - \varepsilon_{y,i+k}) + (\theta_{y,i} - \theta_{y,i+k})(\varepsilon_{x,i} - \varepsilon_{x,i+k}) \\
&\qquad + (\varepsilon_{x,i} - \varepsilon_{x,i+k})(\varepsilon_{y,i} - \varepsilon_{y,i+k})\Big] \\
&= (\theta_{x,i} - \theta_{x,i+k})^2 \operatorname{Var}(\varepsilon_{y,i} - \varepsilon_{y,i+k}) + (\theta_{y,i} - \theta_{y,i+k})^2 \operatorname{Var}(\varepsilon_{x,i} - \varepsilon_{x,i+k}) \\
&\qquad + \operatorname{Var}\left[(\varepsilon_{x,i} - \varepsilon_{x,i+k})(\varepsilon_{y,i} - \varepsilon_{y,i+k})\right] \\
&\qquad + 2(\theta_{x,i} - \theta_{x,i+k})(\theta_{y,i} - \theta_{y,i+k}) \operatorname{Cov}(\varepsilon_{x,i} - \varepsilon_{x,i+k}, \varepsilon_{y,i} - \varepsilon_{y,i+k}) \\
&\qquad + 2(\theta_{x,i} - \theta_{x,i+k}) \operatorname{Cov}(\varepsilon_{y,i} - \varepsilon_{y,i+k}, (\varepsilon_{x,i} - \varepsilon_{x,i+k})(\varepsilon_{y,i} - \varepsilon_{y,i+k})) \\
&\qquad + 2(\theta_{y,i} - \theta_{y,i+k}) \operatorname{Cov}(\varepsilon_{x,i} - \varepsilon_{x,i+k}, (\varepsilon_{x,i} - \varepsilon_{x,i+k})(\varepsilon_{y,i} - \varepsilon_{y,i+k})).
\end{aligned}
$$

Each individual term requires some calculation, but we find that

$$
\operatorname{Var}(\varepsilon_{x,i} - \varepsilon_{x,i+k}) = \operatorname{Var}(\varepsilon_{y,i} - \varepsilon_{y,i+k}) = 2,
$$

$$
\operatorname{Cov}\left(\varepsilon_{x,i} - \varepsilon_{x,i+k}, (\varepsilon_{x,i} - \varepsilon_{x,i+k})(\varepsilon_{y,i} - \varepsilon_{y,i+k})\right) = \mathbb{E}\left[(\varepsilon_{x,i} - \varepsilon_{x,i+k})^2(\varepsilon_{y,i} - \varepsilon_{y,i+k})\right] = 0,
$$

$$
\operatorname{Cov}\left(\varepsilon_{y,i} - \varepsilon_{y,i+k}, (\varepsilon_{x,i} - \varepsilon_{x,i+k})(\varepsilon_{y,i} - \varepsilon_{y,i+k})\right) = 0, \quad \text{and} \quad \operatorname{Cov}(\varepsilon_{x,i} - \varepsilon_{x,i+k}, \varepsilon_{y,i} - \varepsilon_{y,i+k}) = 2\rho.
$$

Finally, we compute the remaining variance term:

$$
\begin{aligned}
&\operatorname{Var}\left[(\varepsilon_{x,i} - \varepsilon_{x,i+k})(\varepsilon_{y,i} - \varepsilon_{y,i+k})\right] \\
&= \mathbb{E}\left[(\varepsilon_{x,i} - \varepsilon_{x,i+k})^2(\varepsilon_{y,i} - \varepsilon_{y,i+k})^2\right] - \left(\mathbb{E}\left[(\varepsilon_{x,i} - \varepsilon_{x,i+k})(\varepsilon_{y,i} - \varepsilon_{y,i+k})\right]\right)^2 \\
&= \mathbb{E}\left[(\varepsilon_{x,i}^2 + \varepsilon_{x,i+k}^2 - 2\varepsilon_{x,i}\varepsilon_{x,i+k})(\varepsilon_{y,i}^2 + \varepsilon_{y,i+k}^2 - 2\varepsilon_{y,i}\varepsilon_{y,i+k})\right] - 4\rho^2 \\
&= \mathbb{E}[\varepsilon_{x,i}^2\varepsilon_{y,i}^2 + \varepsilon_{x,i}^2\varepsilon_{y,i+k}^2 + \varepsilon_{x,i+k}^2\varepsilon_{y,i}^2 + \varepsilon_{x,i+k}^2\varepsilon_{y,i+k}^2 + 4\varepsilon_{x,i}\varepsilon_{x,i+k}\varepsilon_{y,i}\varepsilon_{y,i+k}] - 4\rho^2 \\
&= 2\kappa_{22} + 2.
\end{aligned}
$$

Combining all of these calculations, we get the simplified form:

$$
\begin{aligned}
\operatorname{Var}\left[(X_i - X_{i+k})(Y_i - Y_{i+k})\right] = {} & 2(\theta_{x,i} - \theta_{x,i+k})^2 + 2(\theta_{y,i} - \theta_{y,i+k})^2 \\
& + 2\kappa_{22} + 2 + 4\rho(\theta_{x,i} - \theta_{x,i+k})(\theta_{y,i} - \theta_{y,i+k}).
\end{aligned}
$$

Lemma 4 *Let $X_i = \theta_{x,i} + \varepsilon_{x,i}$ and $Y_i = \theta_{y,i} + \varepsilon_{y,i}$, where $\mathbb{E}[\varepsilon_{x,i}] = \mathbb{E}[\varepsilon_{y,i}] = 0$, $\mathrm{Var}(\varepsilon_{x,i}) = \mathrm{Var}(\varepsilon_{y,i}) = 1$, and $\mathrm{Cov}(\varepsilon_{x,i}, \varepsilon_{y,i}) = \rho$. Let $\kappa_{12} = \mathbb{E}[\varepsilon_{x,i}\varepsilon_{y,i}^2]$, $\kappa_{21} = \mathbb{E}[\varepsilon_{x,i}^2\varepsilon_{y,i}]$, and $\kappa_{22} = \mathbb{E}[\varepsilon_{x,i}^2\varepsilon_{y,i}^2]$.*

Then for any three different indices i, i', i'', the covariance between lag-k cross-products satisfies:

$$\begin{aligned}
&\mathrm{Cov}\left((X_i - X_{i'})(Y_i - Y_{i'}), (X_{i'} - X_{i''})(Y_{i'} - Y_{i''})\right)\\
=&(\kappa_{22} - \rho^2) - (\theta_{x,i} - \theta_{x,i'})(\theta_{x,i'} - \theta_{x,i''}) - (\theta_{y,i} - \theta_{y,i'})(\theta_{y,i'} - \theta_{y,i''})\\
&- (\theta_{x,i} - \theta_{x,i'})(\theta_{y,i'} - \theta_{y,i''})\rho - (\theta_{y,i} - \theta_{y,i'})(\theta_{x,i'} - \theta_{x,i''})\rho\\
&- (\theta_{x,i} - 2\theta_{x,i'} + \theta_{x,i''})\kappa_{12} - (\theta_{y,i} - 2\theta_{y,i'} + \theta_{y,i''})\kappa_{21}.
\end{aligned}$$

We begin by decomposing X and Y into their signal and noise components:

$$\begin{aligned}
&\mathrm{Cov}\left((X_i - X_{i'})(Y_i - Y_{i'}), (X_{i'} - X_{i''})(Y_{i'} - Y_{i''})\right)\\
=&\mathrm{Cov}((\theta_{x,i} - \theta_{x,i'} + \varepsilon_{x,i} - \varepsilon_{x,i'})(\theta_{y,i} - \theta_{y,i'} + \varepsilon_{y,i} - \varepsilon_{y,i'}),\\
&\quad(\theta_{x,i'} - \theta_{x,i''} + \varepsilon_{x,i'} - \varepsilon_{x,i''})(\theta_{y,i'} - \theta_{y,i''} + \varepsilon_{y,i'} - \varepsilon_{y,i''}))\\
=&\mathrm{Cov}((\theta_{x,i} - \theta_{x,i'})(\varepsilon_{y,i} - \varepsilon_{y,i'}) + (\theta_{y,i} - \theta_{y,i'})(\varepsilon_{x,i} - \varepsilon_{x,i'}) + (\varepsilon_{x,i} - \varepsilon_{x,i'})(\varepsilon_{y,i} - \varepsilon_{y,i'}),\\
&\quad(\theta_{x,i'} - \theta_{x,i''})(\varepsilon_{y,i'} - \varepsilon_{y,i''}) + (\theta_{y,i'} - \theta_{y,i''})(\varepsilon_{x,i'} - \varepsilon_{x,i''}) + (\varepsilon_{x,i'} - \varepsilon_{x,i''})(\varepsilon_{y,i'} - \varepsilon_{y,i''}))
\end{aligned}$$

With some calculation, we reduce this to

$$\begin{aligned}
&(\theta_{x,i} - \theta_{x,i'})(\theta_{x,i'} - \theta_{x,i''})(-1) + (\theta_{x,i} - \theta_{x,i'})(\theta_{y,i'} - \theta_{y,i''})(-\rho) + (\theta_{x,i} - \theta_{x,i'})(-\kappa_{12})\\
&+ (\theta_{y,i} - \theta_{y,i'})(\theta_{x,i'} - \theta_{x,i''})(-\rho) + (\theta_{y,i} - \theta_{y,i'})(\theta_{y,i'} - \theta_{y,i''})(-1) + (\theta_{y,i} - \theta_{y,i'})(-\kappa_{21})\\
&+ (\theta_{x,i'} - \theta_{x,i''})\kappa_{12} + (\theta_{y,i'} - \theta_{y,i''})\kappa_{21} + (\kappa_{22} - \rho^2)\\
=&(\kappa_{22} - \rho^2) - (\theta_{x,i} - \theta_{x,i'})(\theta_{x,i'} - \theta_{x,i''}) - (\theta_{y,i} - \theta_{y,i'})(\theta_{y,i'} - \theta_{y,i''})\\
&- (\theta_{x,i} - \theta_{x,i'})(\theta_{y,i'} - \theta_{y,i''})\rho - (\theta_{y,i} - \theta_{y,i'})(\theta_{x,i'} - \theta_{x,i''})\rho\\
&- (\theta_{x,i} - 2\theta_{x,i'} + \theta_{x,i''})\kappa_{12} - (\theta_{y,i} - 2\theta_{y,i'} + \theta_{y,i''})\kappa_{21}.
\end{aligned}$$